# SKP ENGINEERING COLLEGE

Tiruvannamalai – 606 611

## DEPARTMENT OF

## CSE (CYBER SECURITY)

## B.E – CSE (CYBER SECURITY)

## SB8015 – CYBER SECURITY

**REGULATION - 2021**

**Name : .....**

**Register Number : .....**

**Department : .....**

**Year / Semester .....**

**Subject .....**

# SKP ENGINEERING COLLEGE

## Tiruvannamalai – 606 611

## CERTIFICATE

**Name : .....**

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**Register No :** ☐☐☐☐☐☐☐☐☐☐☐☐☐

Certified that this is the bonafide record of practical work done by the aforesaid student

In the _____Laboratory during the year 20____ - 20____

**Staff-In-Charge** **Head of the Department**

Submission for the Practical Examination held on .....

**Internal Examiner** **External Examiner**

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| Exp.no: 01<br><br>Date: | Create a VPC using Cloud Shell |
|---|---|

**How to start your lab and sign in to the Google Cloud console**

1. Click the **Start Lab** button. On the left is the **Lab Details** panel with the following:

    - Time remaining
    - The **Open Google Cloud console** button
    - The temporary credentials that you must use for this lab
    - Other information, if needed, to step through this lab

    ***Note:*** *If you need to pay for the lab, a pop-up opens for you to select your payment method.*

2. Click **Open Google Cloud console** (or right-click and select **Open Link in Incognito Window**) if you are running the Chrome browser. The **Sign in** page opens in a new browser tab.

    ***Tip:*** You can arrange the tabs in separate, side-by-side windows to easily switch between them.

    ***Note:*** *If the* ***Choose an account*** *dialog displays, click* ***Use Another Account***.

3. If necessary, copy the **Google Cloud username** below and paste it into the **Sign in** dialog. Click **Next**.

"Google Cloud username"
Copied!

content_copy

You can also find the **Google Cloud username** in the **Lab Details** panel.

4. Copy the **Google Cloud password** below and paste it into the **Welcome** dialog. Click **Next**.

"Google Cloud password"
Copied!

content_ copy

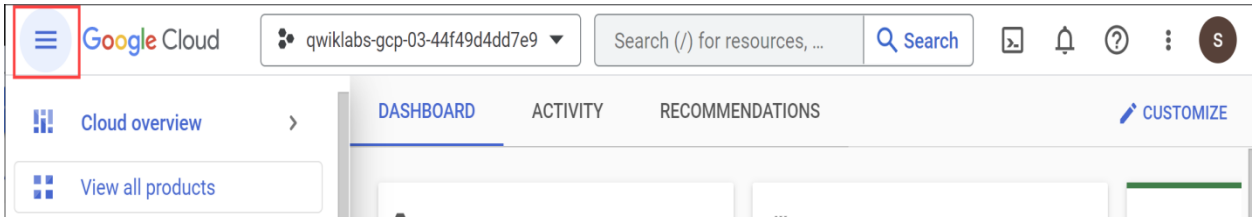
You can also find the **Google Cloud password** in the **Lab Details** panel.

***Important:*** *You must use the credentials the lab provides you. Do not use your Google Cloud account credentials.* ***Note:*** *Using your own Google Cloud account for this lab may incur extra charges.*

5. Click through the subsequent pages:
    - Accept the terms and conditions
    - Do not add recovery options or two-factor authentication (because this is a temporary account)
    - Do not sign up for free trials

After a few moments, the Console opens in this tab.

***Note:*** *You can view the menu with a list of Google Cloud Products and Services by clicking the* ***Navigation menu*** *at the top- left*




**Activate Cloud Shell**

Cloud Shell is an online development and operations environment accessible anywhere with your browser. Cloud Shell provides command-line access to your Google Cloud resources.

1. Click **Activate Cloud Shell** ( ) at the top right of the Google Cloud console. You may be asked to click **Continue**.

After Cloud Shell starts up, you'll see a message displaying your Google Cloud Project ID for this session:

```
Your Cloud Platform project in this session is set to YOUR_PROJECT_ID
```

The command-line tool for Google Cloud, gcloud,comes pre-installed on Cloud Shell and supports tab-completion. In order to access Google Cloud, you'll have to first authorize gcloud.

2. List the active account name with this command:

g cloud auth list
Copied!

content_ copy

3. A pop-up will appear asking you to **Authorize Cloud Shell**. Click **Authorize**.

4. Your output should now look like this:

5. **Output:**

```
ACTIVE: *
ACCOUNT: student-01-xxxxxxxxxxxx@qwiklabs.net
To set the active account, run:
  $ gcloud config set account `ACCOUNT`
```

5. List the project ID with this command:

```
gcloud config list project
```

Copied!

content_ copy

**Example output:**

```
[core]
project = qwiklabs-gcp-44776a13dea667a6
```

**Note:** For full documentation of gcloud, in Google Cloud, refer to the gcloud CLI overview guide.

## Task 1. Create a network

A network forms the basis of communication between devices. You'll need to first create a network for your test environment before you can begin testing security functionality, experimenting with configurations, or building a proof-of-concept for security tools in your role at Cymbal Bank. Here, you'll use software-defined networking to easily set up a network in Google Cloud.

There are two types of VPC networks you can choose to create depending on your subnet requirements. You can choose to create an auto mode or a custom mode VPC network. An auto mode VPC automatically creates a subnet in each region for you while a custom mode VPC provides you with the control to manually create subnets. Each new network that you create must have a unique name within the same project. You can create up to four additional networks in a project.

In this task, you'll create an initial custom mode VPC network.

1. Copy the following command into the Cloud Shell terminal:

***Note:*** *When copying the command and placing it in the Cloud Shell terminal, be sure to note that each command option beginning with "--" needs to be on its own line, and that entering the information incorrectly will result in having to fix the network.*

```
gcloud compute networks create labnet --subnet-mode=custom
```

Copied!

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This command creates a custom mode network called labnet.

2. Press **ENTER**.

Although you don't need to memorize this command, the following breaks it down to help you better understand the syntax:

- gcloud invokes the Cloud SDK gcloud command line tool.
- compute is one of the groups available in gcloud. It lets you create and configure Compute Engine resources and forms part of a nested hierarchy of command groups.
- networks is a subgroup of compute with its own specialized commands. It lets you list, create, and delete Compute Engine networks.
- create is the action to be executed on this group.
- labnet is the name of the network you're creating.
- --subnet-mode=custom is the flag that specifies the type of VPC you are creating, in this instance custom.

The output should list the labnet network you created:

```
NAME: labnet
SUBNET_MODE: CUSTOM
BGP_ROUTING_MODE: REGIONAL
IPV4_RANGE:
GATEWAY_IPV4:
```

Click **Check my progress** to verify that you have completed this task correctly.

Create a network

Check my progress

## Task 2. Create a subnet

In this task, you'll create a subnet within the newly created custom mode VPC network. Configuring subnets is a network management best practice. For test environments, subnets allow you to split your VPC into logical segments to improve the organization of cloud resources, to improve network performance, and to improve security.

When you create a subnet, its name must be unique in that project for that region, even across networks. The same name can appear twice in a project as long as each one is in a different region. Additionally, each subnet must have a primary IP address range, which must be unique within the same region in a project.

1. Copy the following command into the Cloud Shell terminal:

```
gcloud compute networks subnets create labnet-sub \
  --network labnet \
  --region "REGION" \
  --range 10.0.0.0/28
```

Copied!

content_copy

This command creates a sub-network called labnet-sub.

2. Press **ENTER**.

Click **Check my progress** to verify that you have completed this task correctly.

Create a subnet

Check my progress

## Task 3. View networks

In this task, you'll list the available networks to ensure that you have successfully created them.

1. Copy the following command into the Cloud Shell terminal:

gcloud compute networks list
Copied!

content_copy

This command lists the networks in your project.

2. Press **ENTER**.

The output should list the default and labnet networks.

The default network was created when the project was created. The labnet network was created by the gcloud command you ran earlier.

What is the subnet mode of the labnet network you created?

Custom

Default

Auto

None of these options

Submit

## Task 4. List subnets

In this task, you'll list all subnets within the networks of your project.

You can either list all subnets in all networks in your project, or you can show only the subnets for a particular network or region. Auditing subnets ensures that the network is properly secured, and helps identify any misconfigurations or potential security vulnerabilities in your VPCs, such as subnets that might be unintentionally exposed to the public internet.

1. Copy the following command into the Cloud Shell terminal:

```
gcloud compute networks subnets list --network=labnet
```

Copied!

content_copy

This command lists the subnets in the labnet network.

2. Press **ENTER**.

What is the name of the subnet in the labnet network?

labnet

default

labnet-sub

None of these options

Submit

## Conclusion

Great work! By completing this lab activity, you now have hands-on experience in setting up a test VPC network and subnet. This is the first step of creating a test environment which will help you to eventually secure the production environment that will need to protect company data. Thereafter, you were able to confirm the network and subnet's successful creation.

Through observing the network and its subnetworks in the testing environment, you can gather significant data for research. This data is highly beneficial when configuring and creating security plans for the production environment.

| Exp. no: 02<br><br>Date: | Use reports to remediate findings |
|---|---|

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"Google Cloud username"
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content_copy

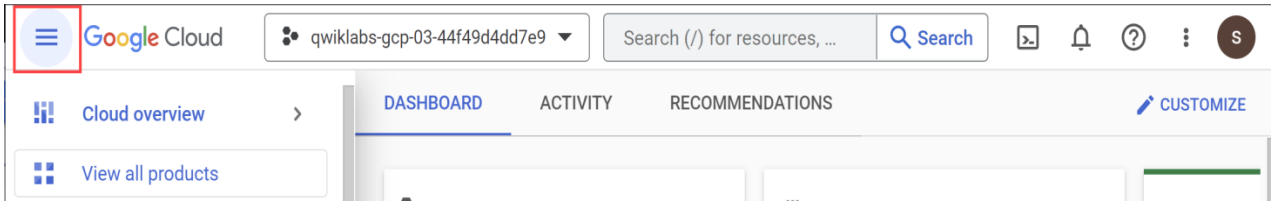
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## Task 1. Identify the vulnerabilities with Security Command Center (SCC)

In this task, you'll use the Security Command Centre (SCC) to check the compliance status of your project, and identify the high and medium risk vulnerabilities that need to be remediated.

1. In the Google Cloud console, from the **Navigation menu** (☰), select **Security** > **Overview**. The Security Command Center Overview page opens.
2. In the **Security Command Center** menu, click **Vulnerabilities**. The Vulnerabilities page opens.

There are many active vulnerabilities listed. You can use the filter to search for the specified findings using the **Module ID**. You will focus on the following active findings listed for your storage bucket:

- **Public bucket ACL (PUBLIC_BUCKET_ACL)**: This entry indicates that there is an Access Control List (ACL) entry for the storage bucket that is publicly accessible which means that anyone on the internet can read files stored in the bucket. This is a high-risk security vulnerability that needs to be prioritized for remediation.

- **Bucket policy only disabled (BUCKET_POLICY_ONLY_DISABLED)**: This entry indicates that uniform bucket-level permissions are not enabled on a bucket. Uniform bucket-level access provides a way to control who can access Cloud Storage buckets and objects, simplifying how

you grant access to your Cloud Storage resources. This is a medium-risk vulnerability that must also be remediated.

- **Bucket logging disabled (BUCKET_LOGGING_DISABLED)**: This entry indicates that there is a storage bucket that does not have logging enabled. This is a low-risk vulnerability that you are not required to remediate in this scenario.

  ***Note:** If the **Public bucket ACL** or **Bucket policy only disabled** are not listed or don't display any active findings, you may have to wait a few minutes and refresh. Wait until these vulnerabilities display active findings before continuing.*

  **Next**, run a compliance report that confirms the vulnerability issues.

3. In the **Security Command Center** menu, click **Compliance**. The Compliance page opens.
4. In the **Google Cloud compliance standards** section, click **View details** in the **CIS Google Cloud Platform Foundation 2.0** tile. The CIS Google Cloud Platform Foundation 2.0 report opens.
5. Click on the **Findings** column to sort the findings and display the active findings at the top of the list.

Which of the following rules in the report have active findings for the Cloud Storage bucket? Select all that apply.

☐ VMs should not be assigned public IP addresses

☐ Bucket policy only should be Enabled

☐ Firewall rules should not allow connections from all IP addresses on TCP or SCTP port 22

☐ Cloud Storage buckets should not be anonymously or publicly accessible

Submit

## Task 2. Remediate the security vulnerabilities

In this task, you'll remediate the security vulnerabilities identified in the previous task. Then, you'll check the security status of the Cloud Storage bucket in the report to confirm that the issues have been remediated.

1. In the Google Cloud console, from the **Navigation menu** (☰), select **Cloud Storage** > **Buckets**.
2. Under the **Filter** section, click the **Name** link of the bucket for your project (BUCKET_NAME). The Bucket details page opens.
3. Click the **Permissions** tab. The **Permissions** section lists all the permissions provided for the bucket.

**First,** remove the public access to the Cloud Storage bucket.

4. Under the **Permissions** section, click the **View by Roles** tab.
5. Expand the **Storage Object Viewer** role and select the checkbox for **allUsers.**
6. Click **Remove Access**.
7. A pop-up will appear asking you to confirm the access removal. Ensure **Remove allUsers from the role Storage Object Viewer on this resource** is selected and click **Remove**.

**Next**, switch the access control to uniform. This will enforce a single (uniform) set of permissions for the bucket and its objects.

8. On the **Access control** tile, click **Switch to uniform**.
9. On the **Edit access control** dialog, select **Uniform**.
10. Click **Save**.

**Finally**, run a compliance report to confirm that the vulnerability issues have been remediated.

11. In the Google Cloud console, from the **Navigation menu** (☰), select **Security** > **Compliance**.
12. In the **CIS Google Cloud Platform Foundation 2.0** tile, and click **View details** to view the report again.

The number of active findings for the **Cloud Storage buckets should not be anonymously or publicly accessible** and **Bucket policy only should be Enabled** rules should now be **0**. This indicates that the **Public bucket ACL** and **Bucket policy only disabled** vulnerabilities for the Cloud Storage bucket have been remediated.

***Note:** If the active findings for the **Public bucket ACL** or **Bucket policy only disabled** don't display as **0** (zero) after you have successfully remediated the vulnerabilities, you may have to wait a few minutes and refresh.*

Click **Check my progress** to verify that you have completed this task correctly.

Remediate the security vulnerabilities

Check my progress

## CONCLUSION

Throughout this lab, you have gained practical experience in identifying and prioritizing threats using the Security Command Center. You also remediated the vulnerabilities identified for your project, and generated a report to confirm that the vulnerabilities have been remediated.

By remediating the vulnerabilities and ensuring the compliance status of the Cloud Storage bucket, you've helped your organization to prevent data breaches, unauthorized access, and data loss.

| Exp. no: 03<br><br>Date: | Create a role in Google Cloud IAM |
|---|---|

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2. Click **Open Google console** (or right-click and select **Open Link in Incognito Window**) if you are running the Chrome browser. The lab **Sign in** page opens in a new browser tab.

    ***Tip:*** You can arrange the tabs in separate, side-by-side windows to easily switch between them.

    ***Note:*** *If the **Choose an account** dialog displays, click **Use Another Account**.*

3. If necessary, copy the **Google Cloud username 1** below and paste it into the **Sign in** dialog. Click **Next**.

"Google Cloud username 1"
Copied!

content_copy

You can also find the **Google Cloud username 1** in the **Lab Details** panel.

4. Copy the **Google Cloud password** below and paste it into the **Welcome** dialog. Click **Next**.

"Google Cloud password"
Copied!

content_copy

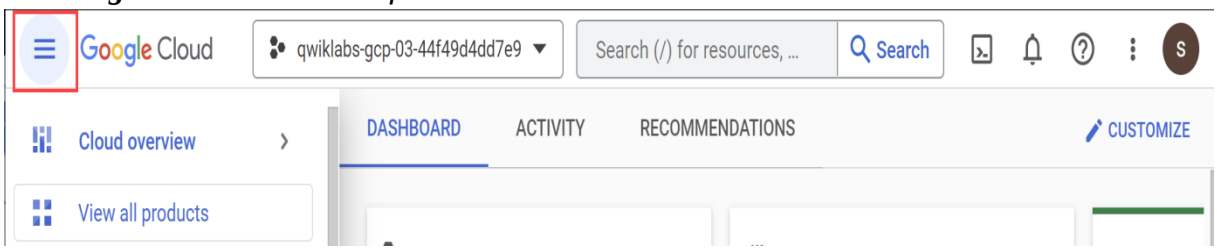
You can also find the **Google Cloud password** in the **Lab Details** panel.

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    - Do not add recovery options or two-factor authentication (because this is a temporary account)
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After a few moments, the Cloud Console opens in this tab.

***Note:*** *You can view the menu with a list of Google Cloud Products and Services by clicking the* ***Navigation menu*** *at the top-* left




## Task 1. Create a custom role

Applying the principle of least privilege is integral to IAM. It ensures that users are only given the permissions they need to perform their tasks. Custom roles provide a way to tailor permissions to an organization's needs, making sure that users do not have broad and excessive permissions.

In this task, you'll create a custom role for the audit team at Cymbal. You'll then grant the custom role restricted access for viewing the database contents.

1. In the Google Cloud console, in the **Navigation menu** (☰), click **IAM & Admin** > **Roles**. The **Roles** page opens.

2. On the Explorer bar, located near the top of the **Roles** page, click **+ Create Role**.

3. In the **Create Role** dialog, specify the following settings and leave the remaining settings as their defaults:

4. Click the **+ Add permissions**. The **Add permissions** dialog box opens.

5. In the **Filter permissions by role** field, type **Firebase Realtime**.

6. In the results drop-down field, select the **Firebase Realtime Database Viewer** checkbox.

7. Click **OK**.

8. Under **Filter**, select
   the **firebase.clients.list** and **firebasedatabase.instances.list** checkboxes to add these permissions to the custom role.

9. Click **Add**.

10. In the **Create Role** dialog, click **Create**.

| Property | Value (type or select) |
| --- | --- |
| Title | Audit Team Reviewer |
| Description | Custom role, allowing the audit team to conduct its review activities. This role grants read-only access to Firebase database resources. |
| ID | CustomRole |
| Role launch stage | General Availability |

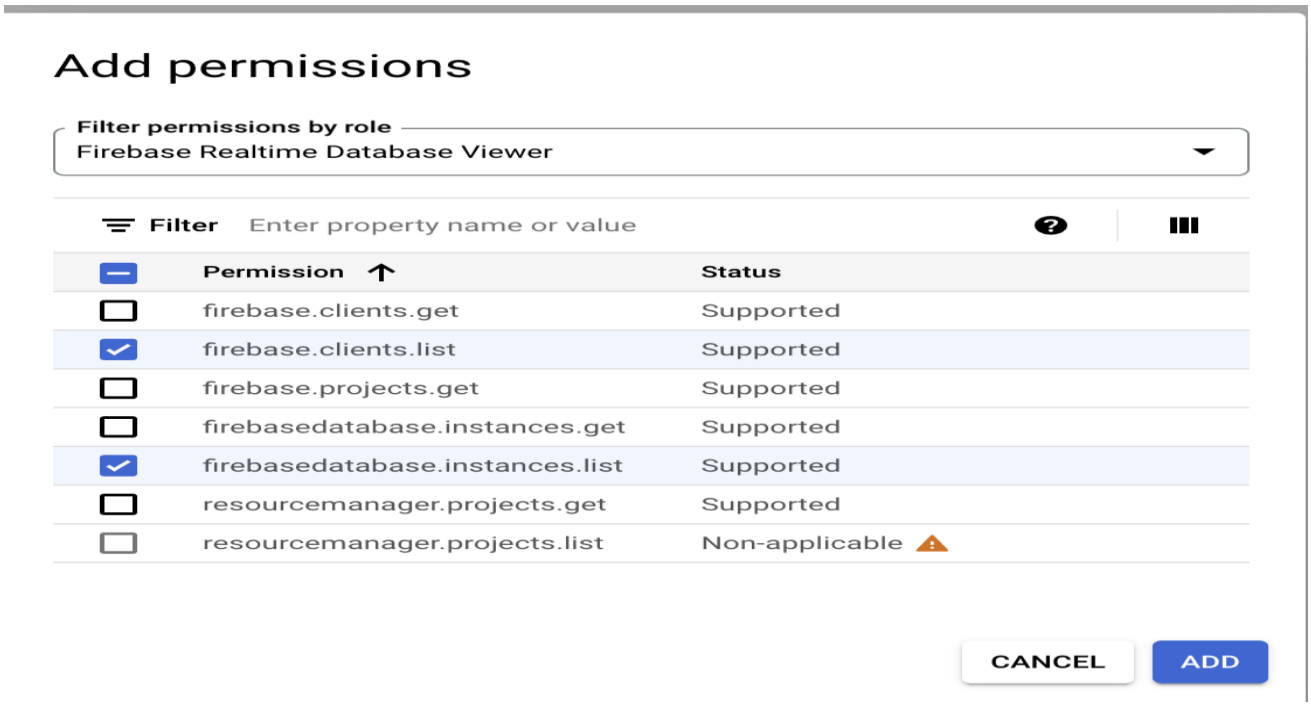
Each custom role can be given a **role launch stage** which reflects the different phases of a role's development, testing, and deployment. These stages help users understand the current state of a role and its suitability for various use cases.

There are several launch stages in Google Cloud. The three primary role launch stages you should know about are:

**Alpha**: Roles in the Alpha stage are typically experimental and may undergo significant changes. They are not recommended for production environments. Users can provide feedback on alpha roles to influence their development.

**Beta**: Roles in the Beta stage are more mature than alpha roles but might still receive updates and improvements based on user feedback. They are considered suitable for certain non-production scenarios but may not be fully stable.

**General Availability (GA)**: Roles that have reached General Availability have undergone thorough development, testing, and refinement. They are considered stable, reliable, and suitable for widespread use in production environments. GA roles have been extensively reviewed and are intended to provide consistent and dependable behavior.




The new role should now be created and added to the existing roles in the project.

Click **Check my progress** to verify that you have completed this task correctly.

Create a custom role

Check my progress

## Task 2. Grant a role to a user

In this task, you'll assign the custom role you created in Task 1 to an existing user.

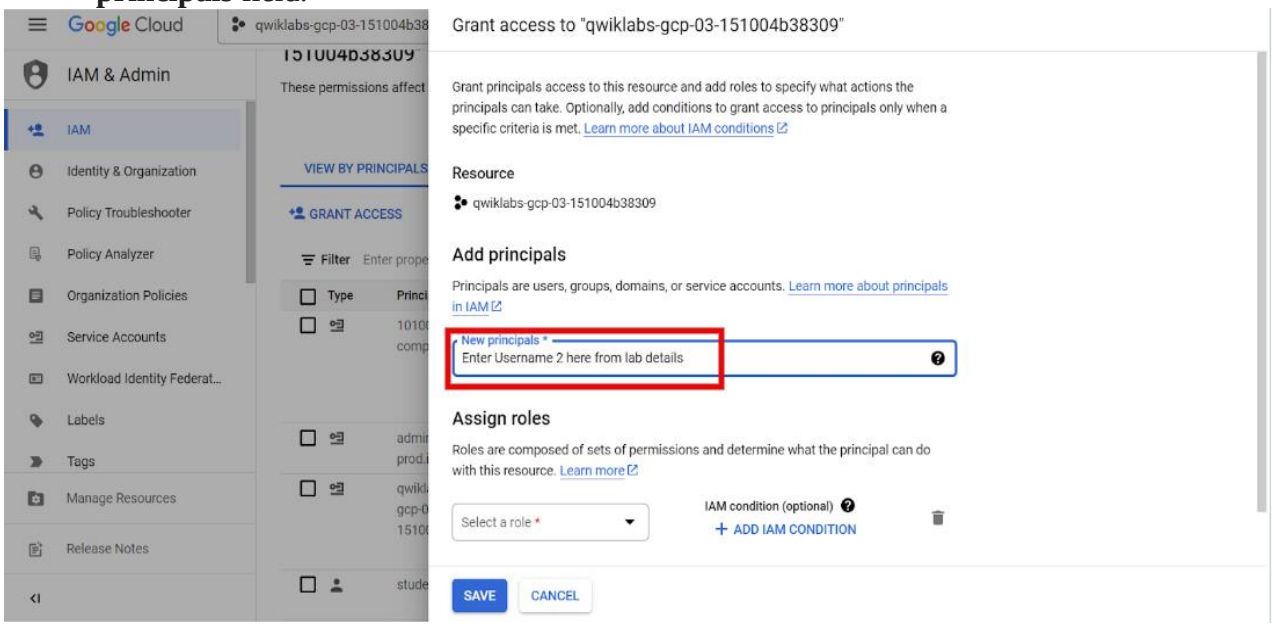
***Note:*** *For the purposes of this lab, you'll grant the new role to* ***Google Cloud username 2*** *provided in the* ***Lab Details*** *panel.*

1. In the Google Cloud console, in the **Navigation menu** (☰), click **IAM & Admin** > **IAM**. The **IAM** page opens.

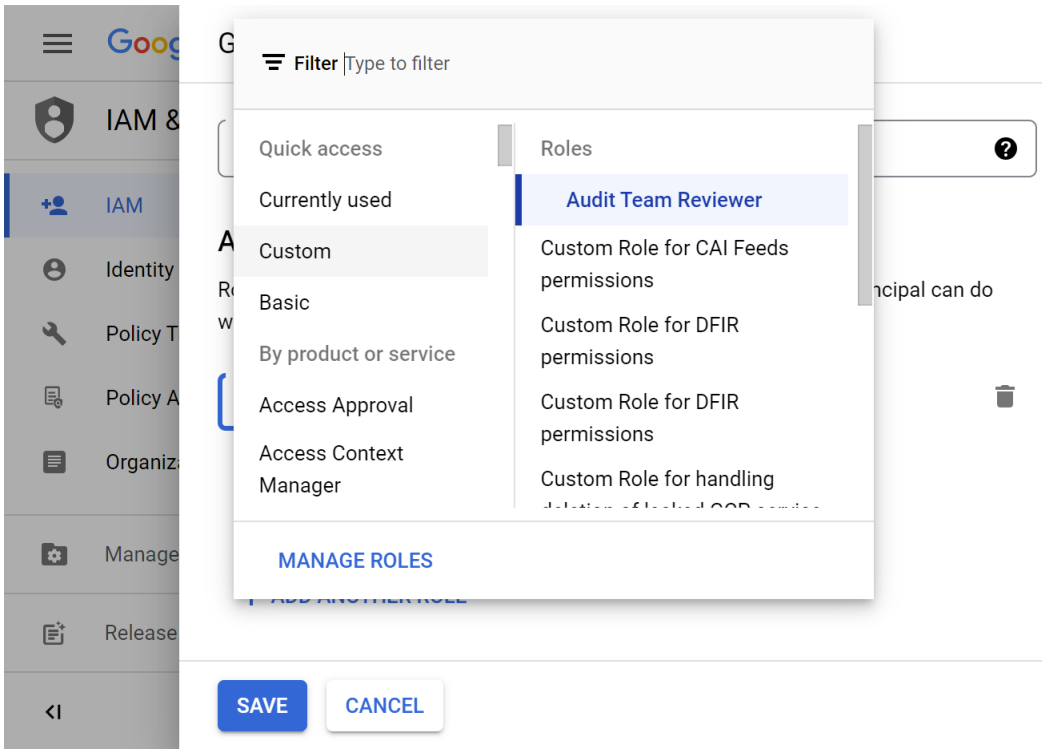
2. On the **View By Principals** tab, click **Grant access**. The **Grant access** dialog window will open.

The **Grant access** dialogue box is a crucial component of the IAM system in Google Cloud. It provides you with the ability to precisely define and manage these permissions for users, groups, and service accounts.

3. Copy the **Google Cloud username 2: Username 2** and paste it into the **New principals** field.




4. Expand the **Select a role** drop-down menu, select **Custom**, and then select **Audit Team Reviewer**. This is the role you created in the previous task.




5. Click **Save**.

The custom role should now be assigned to the user.

Click **Check my progress** to verify that you have completed this task correctly.

Grant a role to a user

Check my progress

## Task 3. Verify the role

So far, you've created a custom role with the appropriate permissions and granted the role to the user. Now, you'll need to check your work to verify that the user has been assigned the role you created. Ensuring that you've correctly configured settings is an integral in part of your workflow as a cloud security analyst.

In this task, you'll use Google Cloud's Policy Analyzer to create a query to check the roles granted to the user.

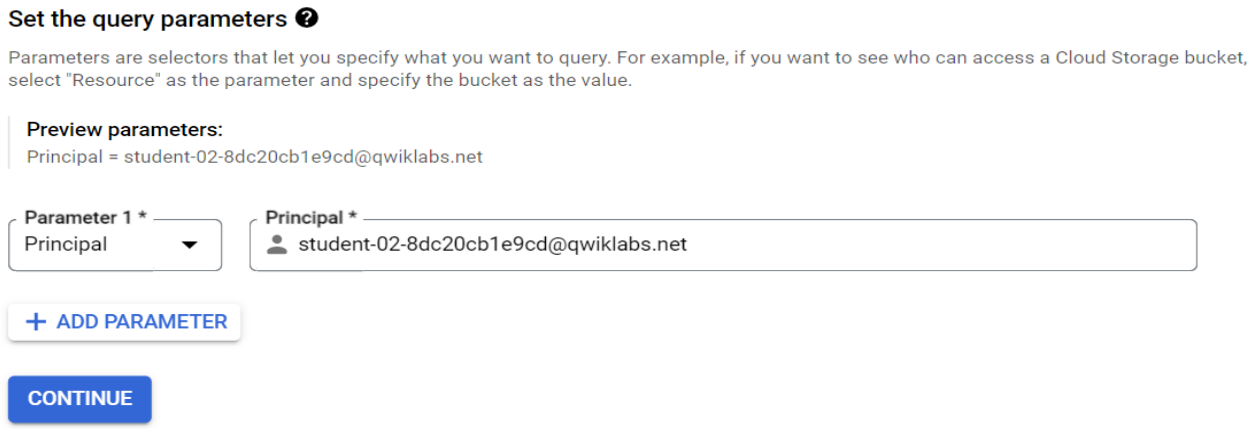
1. In the Google Cloud console, in the **Navigation menu** (☰), click **IAM & Admin** > **Policy Analyzer**. The **Policy Analyzer** page opens.

2. In the **Analyze policies** section, on the **Custom Query** tile, click **Create custom query**. A pop-up may appear at the top left Google Cloud menu (☰) with the text “Click on the

menu anytime to find solutions for your business". Select **Got it** and proceed to the next step.

3. In the **Set the query parameters** section, expand the **Parameter 1** drop-down menu and select **Principal**.

4. Copy the **Google Cloud username 2: Username 2** and paste it into the **Principal** field.




5. Click **Continue**.

6. In the **Advanced options for query results** section, select the **List resources within resource(s) matching your query** checkbox.

7. Click **Analyze** and then select **Run query** in the drop-down menu.

The results should return the role granted to the user. Use the results to answer the following question(s).

Which role has been granted to the user?

BigQuery Admin

Storage Admin

Pub/Sub Admin

Audit Team Reviewer

Submit

## CONCLUSION

Great work! You have successfully utilized IAM to create a custom role, grant access to a user for that role, and verified the permissions within Google Cloud. Cymbal Bank's audit team can now begin working on their database audit using the custom role you created.

IAM defines who has access to which resources based on their role. It is critical for managing digital identities in an organization's environment and will be integral in your work as a cloud security analyst.

By using IAM services, you are well on your way to effectively managing access and permissions to storage resources.

| Exp. no: 04<br><br>Date: | Access a firewall and create a rule |
|---|---|

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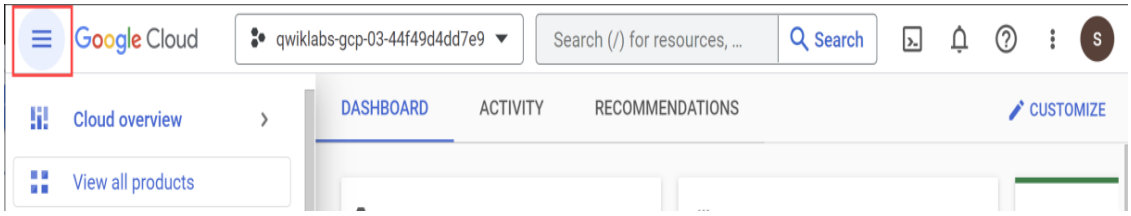
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    - Accept the terms and conditions
    - Do not add recovery options or two-factor authentication (because this is a temporary account)
    - Do not sign up for free trials

After a few moments, the Console opens in this tab.

***Note:*** *You can view the menu with a list of Google Cloud Products and Services by clicking the* ***Navigation menu*** *at the top-left*




## Task 1. Create a firewall rule

In this task, you'll create a firewall rule that allows HTTP and SSH connectivity. You will also specify a target tag for the newly created firewall rule.

In Google Cloud, firewall rules must specify *targets* to define which VM instances they apply to. *Target tags* can be used to apply a firewall rule to a specific group of VMs, helping simplify the management of firewall rules. You'll use target tags to enable this firewall rule to the web server only.

1. In the Google Cloud console, click the **Navigation menu** (☰).
2. Select **VPC Network > Firewall**. The **Firewall policies** page displays.

***Note:*** *If a message is displayed stating that you don't have the required permissions to view the firewall policies inherited by this project, you can disregard it and continue with the next steps.*

3. On the toolbar, click **+ Create Firewall Rule**. The **Create a firewall rule** dialog displays.

4. Specify the following, and leave the remaining settings as their defaults:

| Field | Value |
| --- | --- |
| Name | allow-http-ssh |
| Logs | On |
| Network | vpc-net |
| Targets | Specified target tags |
| Target tags | http-server |
| Source filter | IPv4 ranges |
| Source IPv4 ranges | 0.0.0.0/0 |
| In the **Protocols and ports** section | Select **Specified protocols and ports**<br>Select the **TCP** checkbox<br>In the **Ports** field enter **80, 22** |

5. Click **Create**.

***Note:*** *Wait until the* ***Successfully created firewall rule "allow-http-ssh"*** *message displays before continuing.*

Click **Check my progress** to verify that you have completed this task correctly.

Create a firewall rule

Check my progress

## Task 2. Generate HTTP network traffic

In this task, you'll generate HTTP network traffic to the web server by visiting its external IP address. The network traffic you generate will then be recorded as logs that you can analyze in the Logs Explorer.

**First**, you need to generate network traffic.

1. In the Google Cloud console, click the **Navigation menu** (☰).

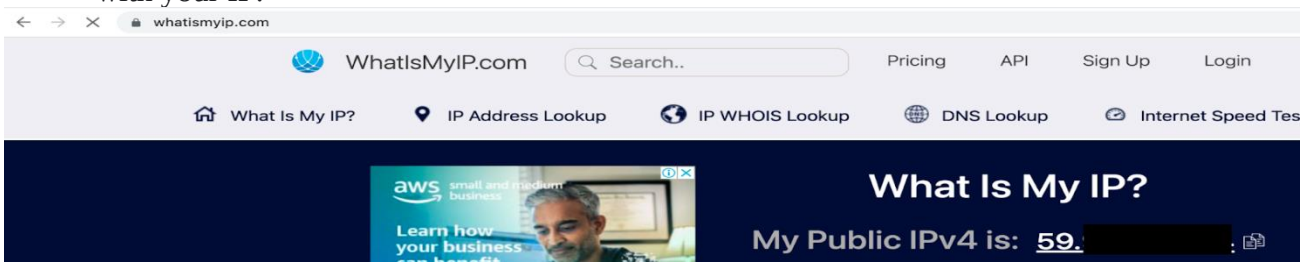
2. Select **Compute Engine > VM instances**. The **VM instances** page opens.

3. For **web-server**, click on the **External IP** link to access the server.

(Alternatively, you can add the **External IP** value to **http://EXTERNAL_IP/** in a new browser window or tab.) A default web page should display.

**Next**, you need to find the IP address of the computer you're using.

4. Access your IP address using the following link whatismyip.com. It will directly reply with your IP.




***Note:*** *Ensure that the IP address only contains numerals (IPv4) and is not represented in hexadecimal (IPv6).*

5. Copy the **IP address** and save it in a notepad. You'll need to use this in the next task.

Click **Check my progress** to verify that you have completed this task correctly.

Generate HTTP network traffic

Check my progress

## Task 3. Analyze the web server Flow Logs

In this task, you'll access and analyze the VPC Flow Logs for the web server using the Logs Explorer.

1. In the Google Cloud console, click the **Navigation menu** (☰).

2. Select **Logging > Logs Explorer**. The **Logs Explorer** page opens. (You may need to expand the **More Products** drop-down menu within the **Navigation** menu and locate Logging under **Operations**.)

3. On the left side of the **Logs Explorer** page, the **Log fields** pane is presented. The **Resource type** and **Severity** sections are available. Under the **Resource type** section, select **Subnetwork**.

Entries from the subnetwork logs will display on the **Query results** pane to the right of the **Log fields** pane.

4. On the **Log fields** pane, in the **Log name** section, select **compute.googleapis.com/vpc_flows** to access the VPC Flow logs for the network. If this option doesn't display, wait a few minutes for this log type to show up.

Once selected, entries from the VPC Flow Logs display on the **Query results** pane.

5. In the **Query** builder at the top of the page, at the end of line 2, press **ENTER** to create a new line.

6. On line 3, enter the following:

jsonPayload.connection.src_ip=YOUR_IP
Copied!

content_copy

Your query should resemble the following:

```
resource.type="gce_subnetwork"
log_name="projects/PROJECT_ID/logs/compute.googleapis.com%2Fvpc_flows"
jsonPayload.connection.src_ip=YOUR_IP
```

7. Replace YOUR_IP with the IP address you saved from Task 2. This query will search for network traffic logs originating from your IP address that you had generated in the previous task.

8. Click **Run query**. The query results should display on the **Query results** pane.

***Note:*** *If the* ***vpc_flows*** *filter option doesn't display or if there are no logs, you might have to wait a few minutes and refresh. If after a couple of minutes, the* ***vpc_flows*** *filter option still doesn't display, navigate to the* ***Compute Engine*** *page and click on the* ***External IP*** *of the* ***web server*** *a few times to generate more traffic and check back on the* ***vpc_flows*** *filter option.*

9. In the **Query results** pane, expand one of the log entries.

10. Within the entry, expand **jsonPayload** by clicking the expand arrow >. Then, expand the **connection** field.

Here you can examine the details about the network connection to the web server:

- **dest_ip** - This is the destination IP address of the web server.
- **dest_port** - This is the destination port number of the web server which is HTTP port 80.
- **protocol** - The protocol is 6 which is the IANA protocol for TCP traffic.
- **src_ip** - This is the source IP address of your computer.
- **src_port** - This is the source port number that's assigned to your computer. According to Internet Assigned Numbers Authority (IANA) standards, this is typically a random port number between 49152-65535.
  After analyzing the details of this log entry, you should notice that the network traffic you generated (on HTTP port 80) was allowed due to the firewall rule **allow-http-ssh** you created previously. This rule allowed incoming traffic on ports 80 and 22.

According to the log entries, what is the IP address of the web server?

0.0.0.0

10.1.3.2

127.0. 0.1

255.255.255.255

Submit

## Task 4. Create a firewall rule to deny HTTP traffic

In this task, you'll create a new firewall rule that denies traffic from port 80.

1. In the Google Cloud console, click the **Navigation menu** (☰).
2. Select **VPC network** > **Firewall**. The Firewall policies page displays.
3. On the toolbar, click **+ Create Firewall Rule**.
4. In the **Create a firewall rule** dialog, specify the following, and leave the remaining settings as their defaults:

| Field | Value |
|---|---|
| Name | deny-http |
| Logs | On |
| Network | vpc-net |
| Action on match | Deny |
| Targets | Specified target tags |
| Target tags | http-server |
| Source filter | IPv4 ranges |

| | |
|---|---|
| Source IPv4 ranges | 0.0.0.0/0 |
| In the **Protocols and ports** section | Select **Specified protocols and ports**<br>Select the **TCP** checkbox<br>In the **Ports** field enter **80** |

5. Click **Create**.

Click **Check my progress** to verify that you have completed this task correctly.

Create a firewall to deny HTTP traffic

Check my progress

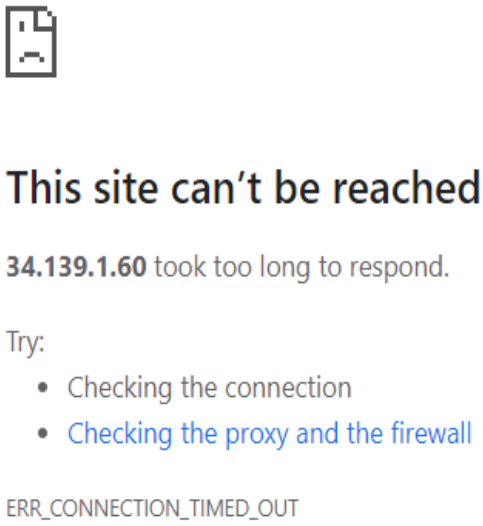
## Task 5. Analyze the firewall logs

In this task, you'll test the **deny-http** firewall rule that you created in the previous task.

**First**, attempt to connect to the web server.

1. Click the **Navigation menu** (☰).
2. Select **Compute Engine > VM instances**. The **VM instances** page opens.
3. For **web-server**, click on the **External IP** link to access the server.

The following error message should display on the page:



This error occurred because of the **deny-http** firewall rule you created in the previous task. To verify this, access the Logs Explorer to analyze the firewall logs for the web server.

4. In the Google Cloud console, click the **Navigation menu** (☰).

5. Select **Logging** > **Logs Explorer**. The **Logs Explorer** page opens. (You may need to expand the **More Products** drop-down menu within the **Navigation** menu and locate Logging under **Operations**.)

6. Under the **Resource type** section, select **Subnetwork**.

7. On the **Log fields** pane, in the **Log name** section, select **compute.googleapis.com/firewall** to access the firewall logs for the network.

8. In the **Query** builder at the top of the page, at the end of line 2, press **ENTER** to create a new line.

9. On line 3, enter the following:

jsonPayload.connection.src_ip=YOUR_IP DENIED
Copied!

content_copy

Replace YOUR_IP with the IP address you saved from Task 2. This query will search for firewall logs that denied your IP address connection to the web server. Your query should resemble the following:

```
resource.type="gce_subnetwork"
log_name="projects/PROJECT_ID/logs/compute.googleapis.com%2Ffirewall"
jsonPayload.connection.src_ip=YOUR_IP DENIED
```

10. Click **Run query**. The query results should display on the Query results pane.

11. In the **Query results** pane, expand one of the log entries.

12. Within the log entry, expand the **jsonPayload** field by clicking the expand arrow >. Then, expand the **connection** field. You can examine the details about the network connection to the web server to verify if the firewall rule was successfully triggered:

- **dest_ip** - This is the destination IP address of the web server which is **10.1.3.2**.
- **dest_port** - This is the destination port number of the web server which is HTTP port **80**.
- **protocol** - The protocol is **6** which is the IANA protocol for TCP traffic.
- **src_ip** - This is the source IP address of your computer.

- **src_port** - This is the source port number that's assigned to your computer.
- **disposition** - This field indicates whether the connection was allowed or denied. Here, it's **denied** which indicates that the connection to the server was denied.

13. Within the log entry, expand the **rule_details** field by clicking the expand arrow >. You can examine the details about the firewall rule. Additionally, you can extract more information from the following fields in the log entry by expanding them:

- **action** - The action taken by the rule, **DENY** in this case.
- **direction** - The rule's traffic direction can be either ingress or egress, here it is **INGRESS** which means the action will apply to incoming traffic.
- **ip_port_info** - The protocol and ports this rule controls.
  The **ip_protocol** and **port_range** lists **TCP port 80**.
- **source_range** - The traffic sources that the firewall rule is applied to. Here it is **0.0.0.0/0**.
- **target_tag** - This lists all the target tags that the firewall rule applies to. Here, it is **http-server**, the target tag you added to the firewall rule in the previous task.
  By examining the details of this firewall log entry, you should notice that the firewall rule **deny-http** you set up to deny HTTP traffic was successfully triggered. This rule denied incoming network traffic on port 80.

Click **Check my progress** to verify that you have completed this task correctly.

Analyze the firewall logs

Check my progress

**CONCLUSION**

You now have practical experience in creating and testing firewall rules for a web server in a cloud environment. By creating firewall rules and analyzing log entries, you have a familiarity with the intricacies of perimeter protection. This is useful for monitoring and analyzing potential security incidents or threats, which is an essential part of a security analyst's role.

You’re well on your way to understanding how to modify firewall rules to ensure maximum network security.

| Exp. no: 05<br><br>Date: | Identify vulnerabilities and remediation techniques |
|---|---|

## How to start your lab and sign in to the Google Cloud console

1. Click the **Start Lab** button. On the left is the **Lab Details** panel with the following:

    - Time remaining
    - The **Open Google Cloud console** button
    - The temporary credentials that you must use for this lab
    - Other information, if needed, to step through this lab

    ***Note:*** *If you need to pay for the lab, a pop-up opens for you to select your payment method.*

2. Click **Open Google Cloud console** (or right-click and select **Open Link in Incognito Window**) if you are running the Chrome browser. The **Sign in** page opens in a new browser tab.

    ***Tip:*** You can arrange the tabs in separate, side-by-side windows to easily switch between them.

    ***Note:*** *If the **Choose an account** dialog displays, click **Use Another Account**.*

3. If necessary, copy the **Google Cloud username** below and paste it into the **Sign in** dialog. Click **Next**.

"Google Cloud username"
Copied!

content_copy

You can also find the **Google Cloud username** in the **Lab Details** panel.

4. Copy the **Google Cloud password** below and paste it into the **Welcome** dialog. Click **Next**.

"Google Cloud password"
Copied!

content_copy

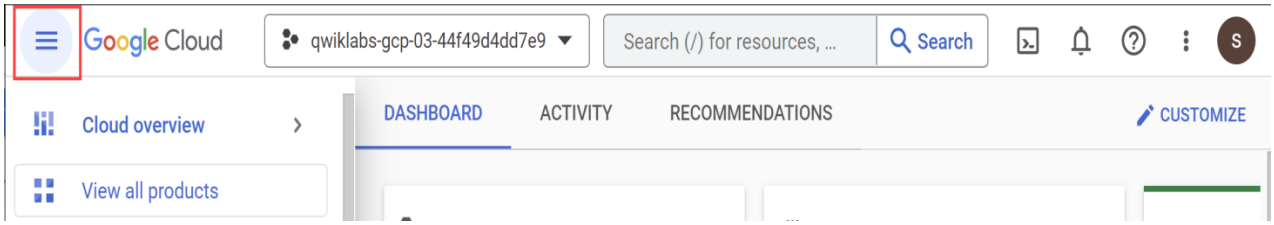
You can also find the **Google Cloud password** in the **Lab Details** panel.

***Important:*** *You must use the credentials the lab provides you. Do not use your Google Cloud account credentials.****Note:*** *Using your own Google Cloud account for this lab may incur extra charges.*

5. Click through the subsequent pages:
    - Accept the terms and conditions
    - Do not add recovery options or two-factor authentication (because this is a temporary account)
    - Do not sign up for free trials

After a few moments, the Console opens in this tab.

***Note:*** *You can view the menu with a list of Google Cloud Products and Services by clicking the* ***Navigation menu*** *at the top-left.*




## Task 1. Launch a virtual machine

In this task, you'll create a static IP address and launch the virtual machine to run the vulnerable application.

1. On the Google Cloud console title bar, click the **Activate Cloud Shell** ( ) icon. If prompted, click **Continue**.
2. Copy the following command into the Cloud Shell terminal:

```
gcloud compute addresses create xss-test-ip-address --region="REGION"
```

Copied!

content_copy

This command creates a static IP address named xss-test-ip-address in the **REGION** region. This static IP will be used for scanning the vulnerable web application.

3. Press **ENTER**.

If prompted, click **Authorize**.

4. Copy the following command into the Cloud Shell terminal:

```
gcloud compute addresses describe xss-test-ip-address \
--region="REGION" --format="value(address)"
```

Copied!

content_copy

This command returns the static IP address you generated.

5. Press **ENTER**.
6. Copy the **IP address** from the output and save it in a notepad. You'll need to use this in a later task.
7. Copy the following command into the Cloud Shell terminal:

```
gcloud compute instances create xss-test-vm-instance --address=xss-test-ip-address --no-service-account \
--no-scopes --machine-type=e2-micro --zone="ZONE" \
--metadata=startup-script='apt-get update; apt-get install -y python3-flask'
```

Copied!

content_copy

This command creates a VM instance to run the vulnerable application.

8. Press **ENTER**.

***Note:*** *The startup script will install* ***python-flask****, a* ***Web Application Framework****, which is used for running a simple Python application. This application demonstrates cross-site scripting (XSS) vulnerability, which is a common web application security vulnerability.*

Click **Check my progress** to verify that you have completed this task correctly.

Launch a virtual machine

Check my progress

## Task 2. Set up and run the vulnerable application

In this task, you'll download and extract the web application files for the vulnerable application, and then deploy the application in the SSH-in-browser.

**First**, you'll create a firewall rule that will allow Web Security Scanner to access the vulnerable application.

1. Copy the following command into the Cloud Shell terminal:

```
gcloud compute firewall-rules create enable-wss-scan \
--direction=INGRESS --priority=1000 \
--network=default --action=ALLOW \
--rules=tcp:8080 --source-ranges=0.0.0.0/0
```

Copied!

content_copy

This command creates a firewall rule that allows access to the web application from any source IP address. This allows the Web Security Scanner to access the vulnerable application and perform a scan.

2. Press **ENTER**.

**Next**, use an SSH connection to connect to the VM instance.

3. In the Google Cloud console, click the **Navigation Menu** (☰).
4. Select **Compute Engine** > **VM instances**.
5. On the **VM instances** page, in the **Connect** column, click on the **SSH** button next to your test instance.

This will open an SSH connection to your VM instance in a new browser window.

6. A pop-up may appear asking you to allow SSH in-browser to connect to VMs. Click **Authorize**.

**Now**, extract the web application files.

7. Copy the following command into the **SSH-in-browser** page (not in Cloud Shell):

```
gsutil cp gs://cloud-training/GCPSEC-ScannerAppEngine/flask_code.tar  . && tar xvf flask_code.tar
```

Copied!

content_copy

This command downloads and extracts the vulnerable web application files.

8. Press **Enter**.

9. **Finally**, copy the following command into the **SSH-in-browser** page:

```
python3 app.py
```

Copied!

content_copy

This command starts the application.

10. Press **ENTER**.

A message should indicate that the application is up and running.

**Note:** Since this is a web application that was installed for use in development, there may be vulnerabilities associated with the configuration file. It is important to test any application prior use on a public facing network.***Note:*** *Do not close the* ***SSH-in-browser*** *page when performing the next task, as the application must continue to run.*

Click **Check my progress** to verify that you have completed this task correctly.

Set up and run the vulnerable application

Check my progress

## Task 3. Access the vulnerable application

In this task, you'll test your application for a vulnerability known as cross-site scripting (XSS). XSS vulnerabilities can be exploited by malicious scripts, such as HTML code, in content that is then served to web browsers.

1. While the application is running, open a new browser window.
2. Copy the URL below into the browser tab, and replace <YOUR_EXTERNAL_IP> with the static IP address of the VM you saved in a notepad in Task 1:

http://<YOUR_EXTERNAL_IP>:8080
Copied!

content_copy

A Cymbal Bank corporate banking portal with a web form should appear.

3. Copy the following HTML code including the script tags into the web form:

<script>alert('This is an XSS Injection to demonstrate one of OWASP vulnerabilities')</script>
Copied!

content_copy

This code injects an OWASP® vulnerability.




4. Click **POST**.

The injected code displayed a message back to the browser. This action by itself is not malicious, however attackers can introduce malicious code into an exploitable application to either steal data from it or implant malware onto the user's device.

The alert window opens with the following message: “This is an XSS Injection to demonstrate one of OWASP vulnerabilities”.

Which security vulnerability is associated with actions performed in this task?

Cross-Site Scripting (XSS)

SQL Injection

Cross-Site Request Forgery (CSRF)

Ransomware

Submit

## Task 4. Scan the application

In this task, you’ll scan the application for vulnerabilities using the Web Security Scanner.

**First**, enable the Web Security Scanner API.

1. In the Google Cloud console, click the **Navigation menu** (☰).
2. Select **APIs & Services** > **Enabled APIs and services**. The **APIs & Services** page displays.
3. Click **+ Enable APIs and services**.
4. In the search field, type **Web Security Scanner**, and press **ENTER**.
5. Select **Web Security Scanner API**.
6. Click **Enable**.

**Now**, scan the application for vulnerabilities.

7. In the Google Cloud console, click the **Navigation menu > View All Products** (☰).
8. Select **Security** > **Web Security Scanner**.

If the **Web Security Scanner API** is enabled then the Cloud Web Security Scanner page displays the **Scan configs** details.

9. In the **Cloud Web Security Scanner** toolbar, click **+ New scan**.

10. In the **Name** section, name the scan **Cross-Site Scripting scan**.

In the **Starting URLs** section, the **Starting URLs 1** field should be pre-populated with your static IP address.

11. Add a colon and the port number **8080** at the end of the IP address. The **Starting URL 1** should resemble the following:

http://<EXTERNAL_IP>:8080

Copied!

content_copy

12. If present, delete **Starting URL 2**.
13. In the **Excluded URLs** section, verify that **Authentication** is set to **None**, and **Schedule** set to **Never**. Leave all other fields unchanged.
14. Click **Save** to create the scan.
15. Click **Run Scan** to start the scan.
16. Return to the **SSH-in-browser** window.

In the **SSH-in-browser** window, you should view logs being generated as Web Security Scanner tests all possible URLs for potential vulnerabilities.

17. When the scan is complete, return to the Google Cloud console.

***Note:*** *The scan might take 5-10 minutes to complete.*

The **Results** tab should indicate the cross-site vulnerabilities, demonstrating how Web Security Scanner can detect a XSS vulnerability.

The vulnerabilities can also be found in the **Vulnerabilities** tab under the **Security Command Centre**.

Click **Check my progress** to verify that you have completed this task correctly.

Scan the application

Check my progress

## Task 5. Remediate the vulnerabilities

In this task, you'll remediate the application's XSS vulnerability and re-run the application with the new fix.

The recommendation for fixing the current vulnerabilities is to validate and escape untrusted user-supplied data, which also points to the corresponding OWASP® rules.

You will do this by editing the code of the vulnerable application to include lines of code that validate and escape the user-supplied data.

1. Return to the **SSH-in-browser** page connected to your VM instance.

2. Press **CTRL + C** to stop the running application. Alternatively, you can click the **Send key combination** icon on the top right corner of the SSH-in-browser window to input the **CTRL + C** key combination.

**Now**, edit the **app.py** file using the nano editor.

3. Copy the following command into the **SSH-in-browser** page:

```
nano app.py
```

Copied!

content_copy

This command opens the nano code editor.

4. Press **ENTER**.

5. To fix the XSS vulnerability, you validate the output string variable. The output string is the processed output of the user-supplied web form input.

Ensure that the application does not accept user input as HTML code, instead, it will escape special characters supplied by user input. To do this, locate the two lines that set the output string:

```
# output_string = "".join([html_escape_table.get(c, c) for c in input_string])
output_string = input_string
```

Copied!

content_copy

6. Remove the # symbol from the first line, and add it to the beginning of the next line (ensure that you indent your code properly.) The final lines must resemble the following:

```
@app.route('/output')
def output():
  output_string = "".join([html_escape_table.get(c, c) for c in input_string])
  # output_string = input_string
  return flask.render_template("output.html", output=output_string)
```

Copied!

content_copy

7. Press **CTRL + X** to exit nano, then **Y** to save your changes, and then **ENTER** to save your changes.
8. Copy the following command into the **SSH-in-browser** terminal:

```
python3 app.py
```

Copied!

content_copy

This command re-runs the application.

9. Press **ENTER**.

Click **Check my progress** to verify that you have completed this task correctly.

Remediate the vulnerabilities

Check my progress

## Task 6. Re-scan the web application

In this task, you'll re-scan the application to make sure there are no vulnerabilities.

1. Return to the **Cloud Web Security Scanner** page in the Google Cloud console.
2. Click **Run** to re-run the scan.

***Note:*** *The scan might take 5-10 minutes to complete.*

The **Results** tab should now indicate that there are no vulnerabilities found.

Click **Check my progress** to verify that you have completed this task correctly. Be sure you wait until the scan completes to get credit for completing this task.

Re-scan the web application

Check my progress

**CONCLUSION**

Through this lab, you gained practical experience in scanning for application vulnerabilities. You learned the importance of a security analyst's ability to scan for application vulnerabilities, which is essential for helping identify and address potential weaknesses, managing risks, meeting compliance requirements, and ultimately, maintaining a robust security posture to protect an organization's assets and reputation.

By closing security gaps and addressing weaknesses, you can help prevent potential exploitation, minimize the impact of security incidents, and maintain compliance with industry regulations.

In this lab, you completed one of the fundamental aspects of proactive cybersecurity strategies.

| Exp. no: 06<br><br>Date: | Change firewall rules using Terraform and Cloud Shell |
|---|---|

## How to start your lab and sign in to the Google Cloud console

1. Click the **Start Lab** button. On the left is the **Lab Details** panel with the following:

    - Time remaining
    - The **Open Google Cloud console** button
    - The temporary credentials that you must use for this lab
    - Other information, if needed, to step through this lab

    ***Note:*** *If you need to pay for the lab, a pop-up opens for you to select your payment method.*

2. Click **Open Google Cloud console** (or right-click and select **Open Link in Incognito Window**) if you are running the Chrome browser. The **Sign in** page opens in a new browser tab.

    ***Tip:*** You can arrange the tabs in separate, side-by-side windows to easily switch between them.

    ***Note:*** *If the* ***Choose an account*** *dialog displays, click* ***Use Another Account***.

3. If necessary, copy the **Google Cloud username** below and paste it into the **Sign in** dialog. Click **Next**.

"Google Cloud username"
Copied!

content_copy

You can also find the **Google Cloud username** in the **Lab Details** panel.

4. Copy the **Google Cloud password** below and paste it into the **Welcome** dialog. Click **Next**.

"Google Cloud password"
Copied!

content_copy

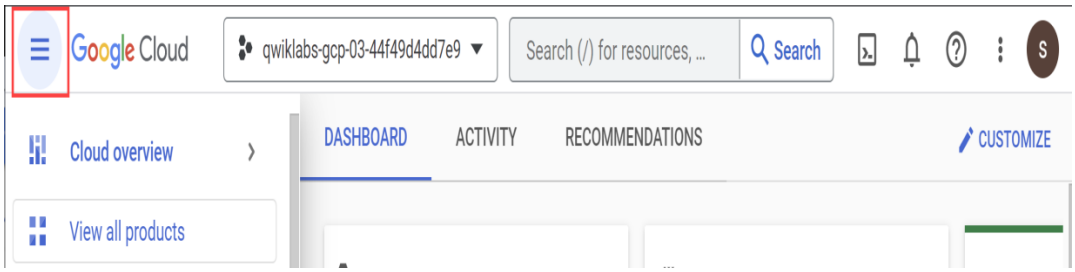
You can also find the **Google Cloud password** in the **Lab Details** panel.

***Important:*** *You must use the credentials the lab provides you. Do not use your Google Cloud account credentials.****Note:*** *Using your own Google Cloud account for this lab may incur extra charges.*

5. Click through the subsequent pages:
    - Accept the terms and conditions
    - Do not add recovery options or two-factor authentication (because this is a temporary account)
    - Do not sign up for free trials

After a few moments, the Console opens in this tab.

***Note:*** *You can view the menu with a list of Google Cloud Products and Services by clicking the* ***Navigation menu*** *at the top-* left




## Task 1. Clone the Terraform repo

In this task, you'll clone the Terraform example repository using the Cloud Shell terminal. The Terraform example contains the configuration file, which you'll use to provision the firewall rules.

1. In the Google Cloud console, click the **Activate Cloud Shell**
2. Click **Continue**.

It should only take a few moments to provision and connect to the Cloud Shell environment.

3. Copy the following command into the Cloud Shell terminal:

```
cloudshell_open --repo_url "https://github.com/terraform-google-modules/docs-examples.git" --print_file "./motd" --dir "firewall_basic" --page "editor" --tutorial "./tutorial.md" --open_in_editor "main.tf" --force_new_clone
```

Copied!

content_copy

This command clones the Terraform example directory.

4. Press **ENTER**.

This command performs the following actions:

- Clones the terraform-google-modules.

- Prints the motd file name.
- Switches to the firewall_basic directory.
- Checks the cloned files, for example tutorial.md.
- Opens main.tf in Cloud Shell Editor.

Once the cloning is complete, you'll be at the ~/cloudshell_open/docs-examples/firewall_basic location in the terminal. Your Cloud Shell prompt should display similar output to the following example:

```
student_01_c2e095df84e2@cloudshell:~/cloudshell_open/docs-examples/firewall_basic (qwiklabs-gcp-04-fde36f013e65)$
```

5. Copy the following command into the Cloud Shell terminal to list the contents of the directory:

```
ls
```

Copied!

content_copy

You should notice that several files in the directory have been downloaded: backing_file.tf, main.tf, motd, and tutorial.md.

6. Copy the following command into the Cloud Shell terminal to analyze the configuration of the firewall rule:

```
cat main.tf
```

Copied!

content_copy

7. Press **ENTER**.

The main.tf file is the configuration file that defines the resources that Terraform will create. Two resources will be created: a firewall rule google_compute_firewall named test-firewall-${local.name_suffix} with rules to allow ICMP and TCP traffic from ports 80, 8080, and 1000-2000 and a VPC network google_compute_network named test-network-${local.name_suffix}. The variable ${local.name_suffix} is a local variable that automatically generates unique names for resources.

Which one of the following protocols are being modified (allow/deny) on the firewall using the Terraform **main.tf** file in **Cloud Shell Editor**?

icmp, web

test-firewall, test-network

web, test-network

icmp, tcp

Submit

## Task 2. Deploy the VPC network and firewall

In this task, you'll deploy a new VPC network and a new firewall rule. This task provides hands-on experience with building a VPC network and subnets.

***Note:*** *Run the following commands in sequence in the Cloud Shell terminal.*

1. Copy the following command into the Cloud Shell terminal.

export GOOGLE_CLOUD_PROJECT=Project ID
Copied!

content_copy

This command sets the project ID.

2. Press **ENTER**.

3. Copy the following command into the Cloud Shell terminal:

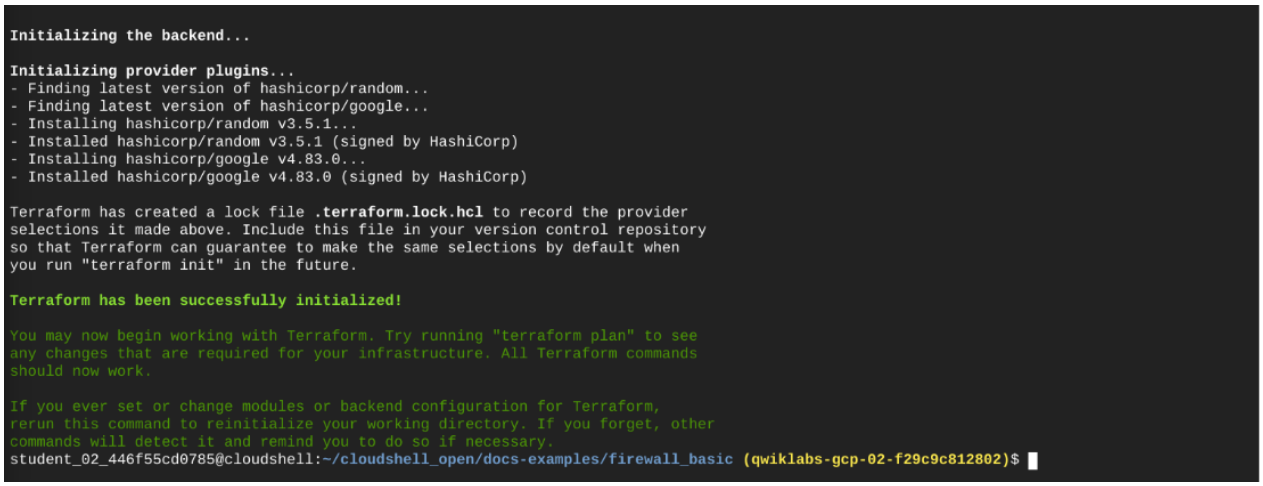
terraform init
Copied!

content_copy

This command initializes the Terraform script.

4. Press **ENTER**.

The output should return a message stating that the Terraform has been successfully initialized. Take a moment to examine the output. You'll notice that Terraform will create a new firewall and VPC network:



5. Once the initialization is complete, copy the following command into the Cloud Shell terminal:

terraform apply
Copied!

content_copy

This command applies the changes and deploys the Terraform script.

6. Press **ENTER**.

***Note:*** *If an* ***Authorize Cloud Shell*** *dialog box appears, click* ***Authorize*** *to grant permission to use your credentials for the gcloud command.*

7. The command prompt will prompt you to **Enter a value**. Type "yes", and press **ENTER**.

This will start creating the VPC network and firewall rules.

Once it's completed, the output should return the following message:

```
Apply complete! Resources: 3 added, 0 changed, 0 destroyed.
```

This means that the VPC and firewall have been successfully deployed.

Click **Check my progress** to verify that you have completed this task correctly.

Check my progress

Check my progress

## Task 3. Verify the deployment of the resources

In this task, you'll verify that the newly created VPC and firewall rules have been successfully deployed.

1. In the Google Cloud console, from the Navigation menu (☰), select **VPC network** > **VPC networks**. The VPC networks page opens.
2. You should notice two VPC networks, **default** and the newest one you just created, **test-network**. Click **test-network** to access the VPC network details.
3. Click **Firewalls**. Use the expand arrow to expand **vpc-firewall-rules**. Under **Protocols and ports** and **Action** you should notice the firewall rules are the same rules as defined in the configuration file: **Allow** and **tcp:80**, **1000-2000**, **8080 icmp**.

***Note:*** *To ensure that resource names are unique, both the* ***test-network*** *and* ***test-firewall*** *names will be dynamically appended with a unique identifier. For example,* ***test-network-curly-penguin****. This unique identifier is generated automatically by the* ***${local.name_suffix}*** *local variable, which is defined in the configuration file. This helps prevent resource naming conflicts and ensures the proper organization of infrastructure components.*

## CONCLUSION

You've successfully built a VPC network and subnet using Terraform and the Cloud Shell. This lab provides the foundation to developing advanced automated solutions that can be given to system administrators to use with Terraform.

By creating the VPC network and firewall, you have gained a better understanding of how it enables you to automate the process of provisioning and modifying firewall rules. This helps establish consistency across various environments, while also helping reduce the chance of human error.

| Exp. no: 07<br><br>Date: | Create symmetric and asymmetric keys |
|---|---|

## How to start your lab and sign in to the Google Cloud console

1. Click the **Start Lab** button. On the left is the **Lab Details** panel with the following:

    - Time remaining
    - The **Open Google Cloud console** button
    - The temporary credentials that you must use for this lab
    - Other information, if needed, to step through this lab

    ***Note:*** *If you need to pay for the lab, a pop-up opens for you to select your payment method.*

2. Click **Open Google Cloud console** (or right-click and select **Open Link in Incognito Window**) if you are running the Chrome browser. The **Sign in** page opens in a new browser tab.

    ***Tip:*** You can arrange the tabs in separate, side-by-side windows to easily switch between them.

    ***Note:*** *If the* ***Choose an account*** *dialog displays, click* ***Use Another Account***.

3. If necessary, copy the **Google Cloud username** below and paste it into the **Sign in** dialog. Click **Next**.

"Google Cloud username"
Copied!

content_copy

You can also find the **Google Cloud username** in the **Lab Details** panel.

4. Copy the **Google Cloud password** below and paste it into the **Welcome** dialog. Click **Next**.

"Google Cloud password"
Copied!

content_copy

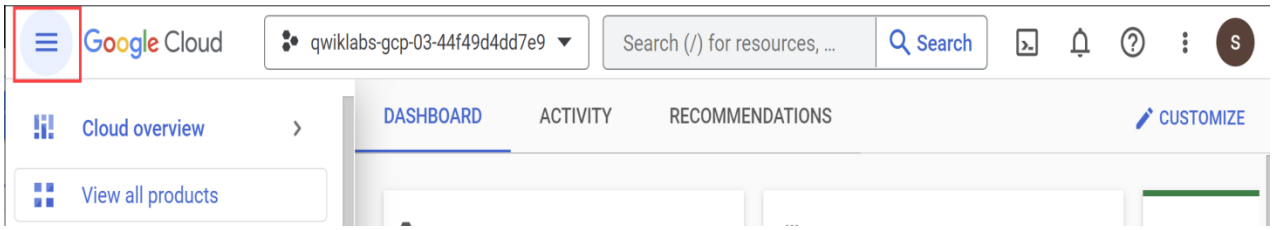
You can also find the **Google Cloud password** in the **Lab Details** panel.

***Important:*** *You must use the credentials the lab provides you. Do not use your Google Cloud account credentials.****Note:*** *Using your own Google Cloud account for this lab may incur extra charges.*

5. Click through the subsequent pages:
    - Accept the terms and conditions
    - Do not add recovery options or two-factor authentication (because this is a temporary account)
    - Do not sign up for free trials

After a few moments, the Console opens in this tab.

***Note:*** *You can view the menu with a list of Google Cloud Products and Services by clicking the* ***Navigation menu*** *at the top-left*




## Task 1. Create a symmetric key

In this task, you'll delve into the intricate process of crafting a symmetric key, complete with considerations for its designated region and the crucial aspect of its protection level. You'll begin by generating a symmetric key with carefully tailored parameters.

1. In the Google Cloud console, click the **Navigation menu** (☰).
2. Select **Security** > **Key Management**.
3. On the **Key Rings** tabbed page, click **+ Create Key Ring**.

**Now**, specify the key details.

4. For **Key ring name**, enter **demo-key-ring**.
5. For the **Location type** category, select **Region**.
6. Expand the **Region** drop-down menu, and select **REGION**.

**Note:** Be sure to select the specified region, otherwise you get an error. ***Note:*** *Be sure you select the specified region, otherwise you get an error.*

7. Click **Create**.

8. In the **Name and protection level** category, in the **Key name** field, enter **demo-key**.

   The **Protection level** should be set to **Software** by default, if not, select it now.

9. Click **Continue**. The Key material category expands.

10. For **Key material**, select **Generated key**.

11. Click **Continue**. The Purpose and algorithm category expands.

12. For **Purpose**, select **Symmetric encrypt/decrypt**.

13. Click **Continue**. The Versions category expands.

14. For **Key rotation period**, select **90 days**.

15. For **Starting on**, leave as the default value.

16. Click **Continue**. No additional settings are needed.

17. Click **Create**.

Once the key is created, it can be used for a variety of implementations such as data encryption and decryption.

**Symmetric keys** are commonly used to encrypt sensitive data before storage or transmission. When data needs to be accessed or shared, the same symmetric key is used to decrypt the encrypted content, ensuring that only authorized parties can access the original information.

Click **Check my progress** to verify that you have completed this task correctly.

Create a symmetric key

Check my progress

Task 2. Create an asymmetric key

In this task, you'll create an asymmetric key with specific settings, including that of its algorithm and protection level.

1. In the Google Cloud console, click the **Navigation menu** (☰).

2. Select **Security > Key Management**. The Key Rings tabbed page opens, listing the newly-created key.

3. Under **Name**, click the link for the key you created in the previous task: **demo-key-ring**. The Key ring details page opens.

4. In the **Keys** tabbed page, click **+ Create Key**.

Now, specify the key details.

5. For **Key name**, enter **demo-asymmetric-key**.

6. For **Protection Level**, select **Software**.

7. Click **Continue**. The Key material category expands.

8. For **Key Material**, select **Generated key**.

9. Click **Continue**. The Purpose and algorithm category expands.

10. For **Purpose**, select **Asymmetric decrypt**.

11. For **Algorithm**, leave as the default value.

12. Click **Continue**.

13. For **Versions**, no settings are required.

14. Click **Continue**. No additional settings are needed.

15. Click **Create**.

The asymmetric key for decryption should now be created.

**Asymmetric keys** can also be used for digital signatures. Digital signatures help verify the authenticity and integrity of messages, files, or software, ensuring that they have not been tampered with during transmission. Digital signatures use two keys, one for signing which involves the user's private key, and one for verifying signatures which involves the user's public key. The output of the signature process is called the digital signature.

Click **Check my progress** to verify that you have completed this task correctly.

Create an asymmetric key

Check my progress

## CONCLUSION

Great work! Through this lab activity, you have gained practical experience in creating both symmetric and asymmetric keys, which play a crucial role in ensuring secure data and communication over networks.

Having created both types of keys, you now have a better understanding of their significance in cryptography. Your newfound ability to create these keys allows you to assist customers in securely storing large amounts of data.

| Exp. no: 08<br><br>Date: | Determine the difference between normal activity and an incident |
|---|---|

## How to start your lab and sign in to the Google Cloud console

1. Click the **Start Lab** button. On the left is the **Lab Details** panel with the following:

    - Time remaining
    - The **Open Google Cloud console** button
    - The temporary credentials that you must use for this lab
    - Other information, if needed, to step through this lab

    ***Note:*** *If you need to pay for the lab, a pop-up opens for you to select your payment method.*

2. Click **Open Google Cloud console** (or right-click and select **Open Link in Incognito Window**) if you are running the Chrome browser. The **Sign in** page opens in a new browser tab.

    ***Tip:*** You can arrange the tabs in separate, side-by-side windows to easily switch between them.

    ***Note:*** *If the* ***Choose an account*** *dialog displays, click* ***Use Another Account****.*

3. If necessary, copy the **Google Cloud username** below and paste it into the **Sign in** dialog. Click **Next**.

"Google Cloud username"
Copied!

content_copy

You can also find the **Google Cloud username** in the **Lab Details** panel.

4. Copy the **Google Cloud password** below and paste it into the **Welcome** dialog. Click **Next**.

"Google Cloud password"
Copied!

content_copy

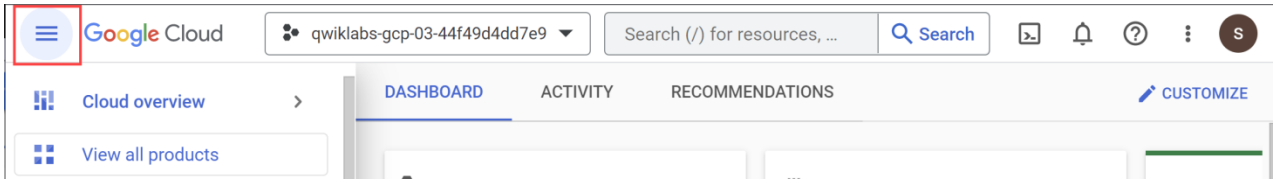
You can also find the **Google Cloud password** in the **Lab Details** panel.

***Important:*** *You must use the credentials the lab provides you. Do not use your Google Cloud account credentials.****Note:*** *Using your own Google Cloud account for this lab may incur extra charges.*

5. Click through the subsequent pages:
    - Accept the terms and conditions
    - Do not add recovery options or two-factor authentication (because this is a temporary account)
    - Do not sign up for free trials

After a few moments, the Console opens in this tab.

***Note:*** *You can view the menu with a list of Google Cloud Products and Services by clicking the* ***Navigation menu*** *at the top-left.*




## Task 1. Grant permissions to an external account

In this task, you'll grant project owner rights to an external gmail account. Granting owner rights to an external account will trigger the Event Threat Detection IAM detectors. Granting project owner rights to an external account is considered anomalous behavior or potentially malicious activity. Event Threat Detection will identify this activity as a threat and generate findings which you'll examine in the upcoming tasks.

1. In the Google Cloud console, in the **Navigation menu** (☰), click **IAM & Admin** > **IAM**. The **IAM** page opens.

On the **View By Principals** tab, note the two student users that have been automatically configured for the qwiklabs.net organization. These two users are also the same users listed in the **Lab details** panel as **Google Cloud username 1** and **Google Cloud username 2**.

These two users have automatically been granted owner roles to the lab project by a service account as part of a normal provisioning process. This will trigger an alert finding or incident because an external principal has an owner role. However, because both users belong to the qwiklabs.net organization this alert is considered normal activity. You will examine this alert finding later.

2. On the **View By Principals** tab, click **Grant Access**. The **Grant access** dialog displays.
3. Under the **Add principals** section, in the **New principals** field, type **bad.actor.demo@gmail.com**.
4. Expand the **Select a role** drop-down menu, select **Basic**, and then select **Owner**.
5. Click **Save**.

You have now assigned the owner role to the external user bad.actor.demo@gmail.com. This will trigger a finding in SCC because this user is outside of the qwiklabs.net organization.

Click **Check my progress** to verify that you have completed this task correctly.

Grant permissions to an external account

Check my progress

## Task 2. Access the Event Threat Detection findings

In this task, you'll access the Event Threat Detection findings in the Security Command Center.

1. In the Google Cloud console, in the **Navigation menu** (☰), click **Security** > **Findings**. The **Findings** page opens.

You should notice three findings with high severities listed in the **Finding query results** panel. In this lab, you'll examine two **Persistence: IAM anomalous grant** findings to determine whether the finding is normal activity or whether it is malicious.

***Note:** If the **Persistence: IAM anomalous grant** findings are not listed, you may have to wait a few minutes and refresh. Wait until both these active findings display before continuing.*

The **Persistence: IAM anomalous grant** indicates that an anomalous IAM grant was detected. This means that a user or service account was granted access to a resource that they should not have had access to. This could be a potential indication of a malicious actor attempting to gain unauthorized access to your environment.

**Next**, filter the findings to display a list of **Persistence: IAM anomalous grant** category findings.

2. In the **Quick filters** panel, in the **Category** section, select the checkbox for the **Persistence: IAM anomalous grant** category.

***Note:** Selecting attributes with quick filters automatically adds them to the query. Notice that the Query preview is updated with the **Persistence: IAM anomalous grant** category you selected. You can locate specific findings or groups of findings by editing the findings query.*

The filter returns two **Persistence: IAM anomalous grant** findings.

3. Click the **Event time** column header to sort the findings in descending order, so that the earliest finding is at the top.

## Task 3. Analyze the findings

In this task, you'll examine these findings to determine which is normal activity and which is a genuine incident.

1. In the **Findings query results** panel, in the **Category** column, click the **Persistence: IAM Anomalous Grant** finding with the earliest event time. The **Persistence: IAM Anomalous Grant** dialog opens on the **Summary** tab, which displays the finding summary.

2. Find the **Principal email** row. This is the user account that granted the owner role to the user. Notice that the service account belongs to the qwiklabs.net organization. With this information, you can establish that this finding represents normal and expected activity.

3. Click the **Source Properties** tab, and expand **properties > sensitiveRoleGrant > members**. Again, the email address listed for **principalEmail** is the user that granted the owner role, and the email address(es) listed for **members** is the user that was granted the owner role.

Which user was granted the owner role in the earliest Persistence: IAM Anomalous Grant finding record?

The external user bad.actor.demo@gmail.com

A Compute Engine service account

None of these options

A user belonging to the qwiklabs.net organization

Submit

**Next**, you'll locate the malicious activity associated with the external user account you had granted access to: bad.actor.demo@gmail.com.

4. Click the close (**X**) button to return to the **Findings** page.
5. In the **Findings query results** panel, in the **Category** column, click on the **Persistence: IAM Anomalous Grant** findings record with the latest event time.
6. Note the value on the **Principal email** row. This is the user account email address that granted the owner role to the user.
7. Click **Source Properties** tab, and expand **properties > sensitiveRoleGrant > members**. You should notice the user account **bad.actor.demo@gmail.com**, which is an external user account. With this information, you can establish that this finding is associated with an unauthorized and malicious actor.

Which user was granted the owner role in the Persistence: IAM Anomalous Grant finding with the latest event time?

The default Compute Engine service account

None of these options

A student user belonging to the qwiklabs.net organization

The external bad.actor.demo@gmail.com user

Submit

Which Persistence: IAM Anomalous Grant finding is a genuine incident?

The finding with the latest event time

The finding with the earlier event time

None of these options

Both findings are genuine incidents

Submit

## Task 4. Access the findings in Cloud Logging

In this task, you'll access the events related to the Security Command Center findings in Cloud Logging.

1. In the Google Cloud console, in the **Navigation menu** (☰) click **Logging > Logs Explorer**. The **Logs Explorer** page opens. (You may need to click **More Products** to expand the **Navigation menu** options and locate **Logging** under **Operations**.)
2. Copy the following query into the **Query** builder at the top of the page:

```
protoPayload.authorizationInfo.permission="resourcemanager.projects.setIamPolicy"
protoPayload.methodName="InsertProjectOwnershipInvite"
```

Copied!

content_copy

This query filters the IAM logs.

3. Click **Run query**. The query results should display on the **Query results** pane.
4. In the **Query results** pane, expand the audit log listed for your project.
5. Click **Expand nested fields**. All the nested fields contained in the log are made visible.

You can now examine the details of the anomalous request event including information such as:

- **authenticationInfo**: The email of the user who made the request.
- **request**: The email identity of the user the anomalous grant was made to.
- **request Metadata**: The IP address of the system where the request was made, the browser user agent of the web browser that was used.

This information can be vital when investigating whether an event is normal activity or an actual threat event.

Which user account made the request to grant the project owner role to the bad.actor@gmail.com user?

A Google Cloud IAM service account

None of these options

An external @gmail.com account

A student user belonging to the qwiklabs.net organization

Submit

## Task 5. Fix the finding

In this task, you'll remediate the malicious **Persistence: IAM Anomalous Grant** finding by removing the project owner role that you had previously assigned to the external user.

1. In the Google Cloud console, in the **Navigation menu** (☰), click **IAM & Admin** > **IAM**. The **IAM** page opens.
2. Next to the **bad.actor.demo@gmail.com** user, click the **Edit principal** ( ✎ ) icon. The **Edit permissions** page opens.
3. Click the **Delete** ( 🗑 ) icon to delete the owner role.
4. Click **Save**.

The policy will be updated, and the owner role removed from the bad.actor.demo@gmail.com user.

Click **Check my progress** to verify that you have completed this task correctly.

Fix the finding

Check my progress

## CONCLUSION

Great work! Through this lab activity, you have gained practical experience in analyzing a security alert to determine whether it is a genuine malicious activity.

You did this by granting permissions to an external user, viewing the Event Threat Detection findings in the Security Command Center, and accessing the findings in Cloud Logging. Finally, you remediated the finding by removing the project owner role from the external user.

As a security analyst, these are skills that can enable you to quickly take steps to contain, mitigate, and remediate any threats.

| Exp. no: 09<br><br>Date: | Explore false positives through incident detection |
|---|---|

## How to start your lab and sign in to the Google Cloud console

1. Click the **Start Lab** button. On the left is the **Lab Details** panel with the following:

    - Time remaining
    - The **Open Google Cloud console** button
    - The temporary credentials that you must use for this lab
    - Other information, if needed, to step through this lab

    ***Note:*** *If you need to pay for the lab, a pop-up opens for you to select your payment method.*

2. Click **Open Google console** (or right-click and select **Open Link in Incognito Window**) if you are running the Chrome browser. The lab **Sign in** page opens in a new browser tab.

    ***Tip:*** You can arrange the tabs in separate, side-by-side windows to easily switch between them.

    ***Note:*** *If the* ***Choose an account*** *dialog displays, click* ***Use Another Account***.

3. If necessary, copy the **Google Cloud username 1** below and paste it into the **Sign in** dialog. Click **Next**.

"Google Cloud username 1"
Copied!

content_copy

You can also find the **Google Cloud username 1** in the **Lab Details** panel.

4. Copy the **Google Cloud password** below and paste it into the **Welcome** dialog. Click **Next**.

"Google Cloud password"
Copied!

content_copy

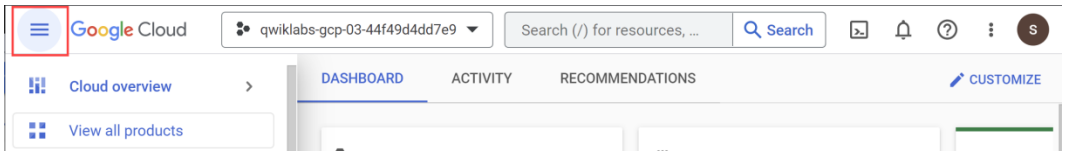
You can also find the **Google Cloud password** in the **Lab Details** panel.

***Important:*** *You must use the credentials from the left panel. Do not use your Google Cloud credentials.****Note:*** *Using your own Google Cloud account for this lab may incur extra charges.*

5. Click through the subsequent pages:
    - Accept the terms and conditions
    - Do not add recovery options or two-factor authentication (because this is a temporary account)
    - Do not sign up for free trials

After a few moments, the Cloud Console opens in this tab.

***Note:*** *You can view the menu with a list of Google Cloud Products and Services by clicking the* ***Navigation menu*** *at the top-lef*




## Task 1. Create a service account

***Note:*** *Make sure you are on the* ***username 1: Google Cloud username 1*** *Google Cloud console.*

In this task, you'll create a service account and grant it permissions sufficient to trigger an anomalous threat finding in SCC.

1. In Google Cloud console, in the **Navigation menu** (☰), click **IAM & Admin** > **Service Accounts**.

2. In the action bar, click **+ Create Service Account**.

3. In the **Service account details** section:

    - In the **Service account name** field, type **test-account**.

      Notice the **Service account ID** automatically populates.

    - Click **Create and Continue**.

      Notice the pop-up message "Service account created".

4. In the **Grant this service account access to project** section, expand the **Select a role** drop-down menu, select **Basic**, and then select **Owner**.

5. Click **Continue**, and then click **Done**.

Notice the **test-account** service account listed in the **Service accounts** list.

Click **Check my progress** to verify that you have completed this task correctly.

Create a service account

Check my progress

## Task 2. Create a JSON authentication key for your service account

***Note:*** *Make sure you are on the* ***username 1: Google Cloud username 1*** *Google Cloud console.*

In this task, you'll create and download a JSON authentication key for the new service account you created in the previous task. You'll then use Cloud Shell to upload that key to your Google Cloud account. This will trigger a threat finding in SCC.

1. Still on the **Service Accounts** page, inline with the **test-account** service account, click **Actions** ( ⋮ ) > **Manage keys**. The **test-account** page opens.

2. In the **Keys** section, click **Add Key** > **Create new key**.

3. In the **Create private key** dialog, set the **Key type** to **JSON**.

4. Click **Create**.

   The console prompts you to download the key to your local device. Once downloaded, you'll use Cloud Shell to upload the key to your Google Cloud (student) account.

5. On your local device, navigate to the key file you just downloaded and rename it **test-account**.

6. In the Google Cloud console, click the Activate Cloud Shell ( ) icon.

7. Click **Continue**.

   It should only take a few moments to provision and connect to the Cloud Shell environment.

8. In the Cloud Shell title bar, click **More** ( ⋮ ) > **Upload** > **Choose Files**.

9. Navigate to and select the file on your local machine, and then in the **Upload** dialog, click **Upload**.

10. Copy the following command into the Cloud Shell terminal:

    ls
    Copied!

    content_copy

This command lists the key file you just uploaded.

11. Press **ENTER**.

In the **test-account** page, in the **Key** list, notice the key you just created with the **Key creation date** as the current date.

Click **Check my progress** to verify that you have completed this task correctly.

Create a JSON authentication key for your service account

Check my progress

## Task 3. Trigger the false positive finding

***Note:*** *Make sure you are on the* ***username 1: Google Cloud username 1*** *Google Cloud console.*

In this task, you'll reconfigure the Cloud Shell environment to use the new **test-account** service account that you created in Task 1. This will trigger a threat finding in SCC. Then, you'll assign excessive permissions to the lab project.

1. Copy the following command into the Cloud Shell terminal:

2. `export PROJECT_ID=$(gcloud info --format='value(config.project)')`
3. `export SA_NAME="test-account@${PROJECT_ID}.iam.gserviceaccount.com"`
   `gcloud auth activate-service-account ${SA_NAME} --key-file=test-account.json`
   Copied!

   content_copy

This command activates the new service account.

2. Press **ENTER**.

3. Copy the following command into the Cloud Shell terminal:

4. gcloud auth list

   Copied!

   content_copy

This command confirms that you activated the service account, and that gcloud is using this service account.

4. Press **ENTER**.

   In the output, the following confirms the service account is active:

   Output:

```
ACTIVE: *
ACCOUNT: test-account@"Google Cloud project ID".iam.gserviceaccount.com
```

5. Copy the following command into the Cloud Shell terminal:

6. export STUDENT2="Google Cloud username 2"
   gcloud projects add-iam-policy-binding $PROJECT_ID --member user:$STUDENT2 --role roles/editor
   Copied!

   content_copy

This command grants the editor role to user 2 so that you can access and remediate the false positive finding in the next task.

6. Press **ENTER**.

Click **Check my progress** to verify that you have completed this task correctly.

Assign excessive permissions to trigger threat detection

Check my progress

## Task 4. Sign in as the second user

You'll need to switch Google Cloud accounts by logging into the Google Cloud console using the second user account provided in the **Lab Details** panel. You will use this user account to perform the remaining tasks.

1. In the Google Cloud console, click on the user icon in the top-right corner of the screen, and then click **Add account**.

2. Navigate back to the **Lab Details** panel, copy the **Google Cloud username 2**: Google Cloud username 2 and **password**. Then, paste the username and password into the Google Cloud console **Sign in** dialog.

## Task 5. View the threat finding in SCC

***Note:** Make sure you are on the **username 2: Google Cloud username 2** Google Cloud console.*

In this task, you'll locate and examine the SCC finding generated by the service Event Threat Detection. This finding is a false positive that was triggered by the activity you generated in Tasks 1-3.

To view the Event Threat Detection finding in SCC:

1. In the **Navigation menu** (☰), click **Security > Findings**.

2. In the **Quick filters** pane, locate the **Category** section, then select **User managed service account key**. If necessary, click **View more** to find it.

   The **Findings query results** panel updates to display only the selected finding category.

3. In the **Findings query results** panel, display the details of the finding by clicking the most recent (see **Event time**) **User managed service account key** in the **Category** column. The details panel for the finding opens and displays the **Summary** tab.

Leave the **User managed service account key** page open to answer the following questions.

1. What is the severity of the alert?

High

Critical

Medium

Low

Submit

2. What is the threat finding class for the alert?

Vulnerability

Misconfiguration

Observation

Threat

Submit

3. When is it important to monitor for threats?

Whenever you access the corporate network on your tablet or smartphone

Whenever your device is on

Whenever your device is connected to the internet

Whenever you device is on site

Submit

4. Which tab in the **User managed service account key** page provides compliance standards, explanation of the threat, and a recommendation on how to handle the threat?

Summary

Source Properties

JSON

Submit

## Task 6. Fix the finding

***Note:*** *Make sure you are on the* ***username 2:*** ***Google Cloud username 2*** *Google Cloud console.*

In this task, you'll remediate the false positive by deleting the JSON authentication key for the **test-account** service account.

1. In Google Cloud console, in the **Navigation menu** (☰), click **IAM & Admin > Service Accounts**.

2. On the **Service accounts** page, click the email address of the **test-account** service account.

3. Click the **Keys** tab.

4. From the list of keys, click the **Delete service account key** ( 🗑 ) icon to delete the key. A pop-up will appear asking you to confirm the action. Click **Delete**.

Click **Check my progress** to verify that you have completed this task correctly.

Delete the key

Check my progress

## Conclusion

You have completed this lab! You used SCC to investigate a false positive and took action to remediate it. As a cloud security analyst, you'll likely encounter false positive alerts. It's important to understand how and why false positive alerts are triggered and how you can take action to remediate them

.

| Exp. no: 10<br><br>Date: | Analyze audit logs using Big Query |
|---|---|

## How to start your lab and sign in to the Google Cloud console

1. Click the **Start Lab** button. On the left is the **Lab Details** panel with the following:

    - Time remaining
    - The **Open Google Cloud console** button
    - The temporary credentials that you must use for this lab
    - Other information, if needed, to step through this lab

    ***Note:*** *If you need to pay for the lab, a pop-up opens for you to select your payment method.*

2. Click **Open Google Cloud console** (or right-click and select **Open Link in Incognito Window**) if you are running the Chrome browser. The **Sign in** page opens in a new browser tab.

    ***Tip:*** You can arrange the tabs in separate, side-by-side windows to easily switch between them.

    ***Note:*** *If the* ***Choose an account*** *dialog displays, click* ***Use Another Account****.*

3. If necessary, copy the **Google Cloud username** below and paste it into the **Sign in** dialog. Click **Next**.

"Google Cloud username"
Copied!

content_copy

You can also find the **Google Cloud username** in the **Lab Details** panel.

4. Copy the **Google Cloud password** below and paste it into the **Welcome** dialog. Click **Next**.

"Google Cloud password"
Copied!

content_copy

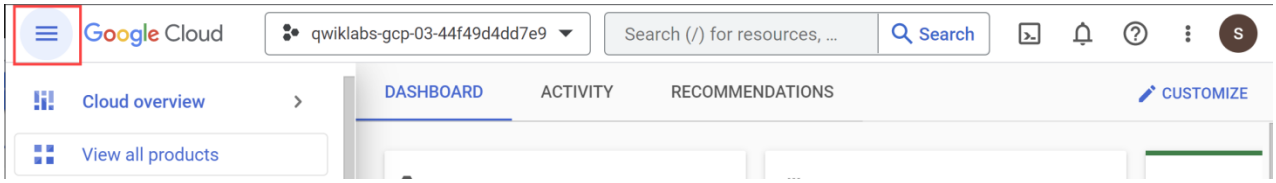
You can also find the **Google Cloud password** in the **Lab Details** panel.

***Important:*** *You must use the credentials the lab provides you. Do not use your Google Cloud account credentials.****Note:*** *Using your own Google Cloud account for this lab may incur extra charges.*

5. Click through the subsequent pages:
    - Accept the terms and conditions
    - Do not add recovery options or two-factor authentication (because this is a temporary account)
    - Do not sign up for free trials

After a few moments, the Console opens in this tab.

***Note:*** *You can view the menu with a list of Google Cloud Products and Services by clicking the* ***Navigation menu*** *at the top-left.*




**Activate Cloud Shell**

Cloud Shell is an online development and operations environment accessible anywhere with your browser. Cloud Shell provides command-line access to your Google Cloud resources.

1. Click **Activate Cloud Shell** ( ) at the top right of the Google Cloud console. You may be asked to click **Continue**.

After Cloud Shell starts up, you'll see a message displaying your Google Cloud Project ID for this session:

```
Your Cloud Platform project in this session is set to YOUR_PROJECT_ID
```

The command-line tool for Google Cloud, gcloud,comes pre-installed on Cloud Shell and supports tab-completion. In order to access Google Cloud, you'll have to first authorize gcloud.

2. List the active account name with this command:

```
gcloud auth list
```

Copied!

content_copy

3. A pop-up will appear asking you to **Authorize Cloud Shell**. Click **Authorize**.

4. Your output should now look like this:

**Output:**

```
ACTIVE: *
ACCOUNT: student-01-xxxxxxxxxxxx@qwiklabs.net
To set the active account, run:
  $ gcloud config set account `ACCOUNT`
```

5. List the project ID with this command:

```
gcloud config list project
```

Copied!

content_copy

**Example output:**

```
[core]
project = qwiklabs-gcp-44776a13dea667a6
```

***Note:*** *For full documentation of gcloud, in Google Cloud, refer to the gcloud CLI overview guide.*

## Task 1. Generate account activity

***Note:*** *Make sure you are on the* ***username 1: Google Cloud username 1*** *Google Cloud console.*

In this task, you'll create and delete cloud resources to generate account activity which you'll access as Cloud Audit Logs.

1. Copy the following commands into the Cloud Shell terminal:

```
gcloud storage buckets create gs://$DEVSHELL_PROJECT_ID

echo "this is a sample file" > sample.txt

gcloud storage cp sample.txt gs://$DEVSHELL_PROJECT_ID

gcloud compute networks create mynetwork --subnet-mode=auto

export ZONE=$(gcloud compute project-info describe \
--format="value(commonInstanceMetadata.items[google-compute-default-zone])")

gcloud compute instances create default-us-vm \
--machine-type=e2-micro \
--zone=$ZONE --network=mynetwork

gcloud storage rm --recursive gs://$DEVSHELL_PROJECT_ID
```

Copied!

content_copy

2. Press **ENTER**.

Click **Check my progress** to verify that you have completed this task correctly.

Generate account activity

Check my progress

## Task 2. Export the audit logs

***Note:*** *Make sure you are on the* ***username 1: Google Cloud username 1*** *Google Cloud console.*

The activity you generated in the previous task was recorded as audit logs. In this task you'll export these logs to a BigQuery dataset for further analysis.

1. In the Google Cloud console, in the **Navigation menu** (☰) click **Logging > Logs Explorer**. The **Logs Explorer** page opens. (You may need to click **More Products** to expand the **Navigation menu** options and locate **Logging** under **Operations**.)
2. When exporting logs, the current filter will be applied to what is exported. Copy the following query into the **Query** builder:

```
logName = ("projects/Project ID/logs/cloudaudit.googleapis.com%2Factivity")
```

Copied!

content_copy

3. Click **Run query**. The query results should display on the **Query results** pane. This query filters for Cloud Audit logs within your project.
4. Under the **Query editor** field, click **More actions > Create sink.** The **Create logs routing sink** dialog opens.

***Note:*** *If your browser window is narrow, the UI may display* ***More*** *instead of* ***More actions****.*

5. In the **Create logs routing sink** dialog, specify the following settings and leave all other settings at their defaults:

| Section | Field: values |
|---|---|
| **Sink details** | Sink name: AuditLogsExport<br>Click **Next.** |

| | |
|---|---|
| **Sink destination** | **Select sink service:** BigQuery dataset<br>**Select BigQuery dataset:** Create new BigQuery dataset.<br>The **Create dataset** dialog opens. |
| **Create dataset** | **Dataset ID:** auditlogs_dataset<br>Click **Create Dataset**.<br>The **Create dataset** dialog closes, and you'll return to the **Sink destination** dialog. |
| **Sink destination** | Click **Next**.<br>Uncheck the **Use Partitioned Tables** checkbox, if it is already selected, and click **Next**. |
| **Choose logs to include in sink** | Notice the pre-filled Build inclusion filter:logName=("projects/[PROJECT ID]/logs/cloudaudit.googleapis.com%2Factivity")<br>Click **Next**.<br>Click **Create Sink**.<br>Return to the **Logs Explorer** page. |

6. In the **Logging** navigation pane, click **Log Router** to view the **AuditLogsExport** sink in the **Log Router Sinks** list.
7. Inline with the **AuditLogsExport** sink, click **More actions** ( ⋮ ) > **View sink details** to view information about the **AuditLogsExport** sink you created. The **Sink details** dialog opens.
8. Click **Cancel** to close the **Sink details** dialog when you're done viewing the sink information.

All future logs will now be exported to BigQuery, and the BigQuery tools can be used to perform analysis on the audit log data. The export does not export existing log entries.

Click **Check my progress** to verify that you have completed this task correctly.

Export the audit logs

Check my progress

## Task 3. Generate more account activity

***Note:*** *Make sure you are on the* ***username 1: Google Cloud username 1*** *Google Cloud console.*

In this task, you'll create and delete cloud resources to generate additional account activity which you'll then access in BigQuery to extract additional insights from the logs.

1. Copy the following commands into the Cloud Shell terminal:

```
gcloud storage buckets create  gs://$DEVSHELL_PROJECT_ID

gcloud storage buckets create  gs://$DEVSHELL_PROJECT_ID-test

echo "this is another sample file" > sample2.txt

gcloud storage cp sample.txt gs://$DEVSHELL_PROJECT_ID-test

export ZONE=$(gcloud compute project-info describe \
--format="value(commonInstanceMetadata.items[google-compute-default-zone])")

gcloud compute instances delete --zone=$ZONE \
--delete-disks=all default-us-vm
```

Copied!

content_copy

These commands generate more activity to view in the audit logs exported to BigQuery.

2. Press **ENTER**.

When prompted, enter **Y**, and press **ENTER**. Notice you created two buckets and deleted a Compute Engine instance.

3. When the prompt appears after a few minutes, continue by entering the following commands into the Cloud Shell terminal:

```
gcloud storage rm --recursive  gs://$DEVSHELL_PROJECT_ID

gcloud storage rm --recursive gs://$DEVSHELL_PROJECT_ID-test
```

Copied!

content_copy

4. Press **ENTER**.

Notice you deleted both buckets.

Click **Check my progress** to verify that you have completed this task correctly.

Generate more account activity

Check my progress

## Task 4. Sign in as the second user

You'll need to switch Google Cloud accounts by logging into the Google Cloud console using the second user account provided in the **Lab Details** panel. You will use this user account to analyze the logs.

1. In the Google Cloud console, click on the user icon in the top-right corner of the screen, and then click **Add account**.
2. Navigate back to the **Lab Details** panel, copy the **Google Cloud username 2: Google Cloud username 2** and **password**. Then, paste the username and password into the Google Cloud console **Sign in** dialog.

## Task 5. Analyze the Admin Activity logs

***Note:*** *Make sure you are on the* ***username 2: Google Cloud username 2*** *Google Cloud console.*

In this task, you'll review the Admin activity logs generated in the previous task. Your goal is to identify and apply filters to isolate logs that may indicate suspicious activity. This will enable you to export this subset of logs and streamline the process of analyzing them for potential issues.

Admin Activity logs record the log entries for API calls or other administrative actions that modify the configuration or metadata of resources. For example, the logs record when VM instances and App Engine applications are created when permissions are changed.

***Note:*** *You can view audit log entries in the Logs Viewer, Cloud Logging, and in the Cloud SDK. You can also export audit log entries to Pub/Sub, BigQuery, or Cloud Storage.*

1. In the Google Cloud console, click the **Navigation menu** (☰).
2. Select **Logging > Logs Explorer**. The **Logs Explorer** page opens. (You may need to expand the **More Products** drop-down menu within the **Navigation** menu and locate Logging under **Operations**.)
3. Ensure that the **Show query** toggle button is activated. This opens the **Query builder** field.
4. Copy and paste the following command into the **Query builder** field. Notice your Google Cloud project ID, project ID in the command.

```
logName = ("projects/"PROJECT_ID"/logs/cloudaudit.googleapis.com%2Factivity")
```

Copied!

content_copy

5. Click **Run query**.

6. In the **Query results**, locate the log entry indicating that a Cloud Storage bucket was deleted, it will contain the **storage.buckets.delete** summary field. Summary fields are included in the log results to highlight important information about the log entry.

This entry refers to **storage.googleapis.com**, which calls the **storage.buckets.delete** method to delete a bucket. The bucket name is the same name as your project id: PROJECT_ID.

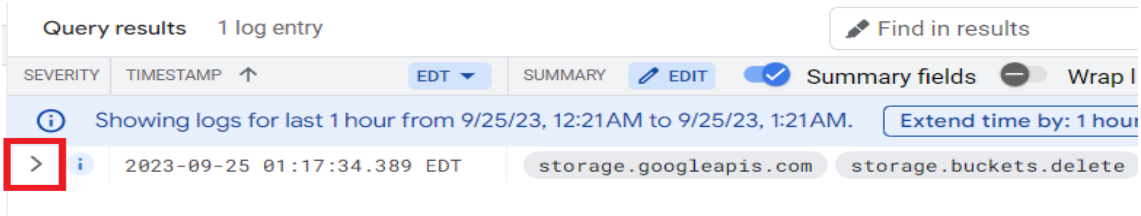
7. Within this entry, click on the **storage.googleapis.com** text, and select **Show matching entries**. The **Query results** should now display only six entries related to created and deleted cloud storage buckets.
8. In the Query editor field, notice the **protoPayload.serviceName="storage.googleapis.com"** line was added to the query builder, this filters your query to entries only matching **storage.googleapis.com**.
9. Within those query results, click **storage.buckets.delete** in one of the entries, and select **Show matching entries.**

Notice another line was added to the **Query builder** text:

```
logName = ("projects/"PROJECT_ID"/logs/cloudaudit.googleapis.com%2Factivity")
protoPayload.serviceName="storage.googleapis.com"
protoPayload.methodName="storage.buckets.delete"
```

The **Query results** should now display all entries related to deleted Cloud Storage buckets. You can use this technique to easily locate specific events.

10. In the **Query results**, expand a **storage.buckets.delete** event by clicking the expand arrow > next to the line:




11. Expand the **authenticatitonInfo** field by clicking the expand arrow > next to the line:




Notice the **principalEmail** field which displays the email address of the user account that performed this action which is the user 1 account you used to generate the user activity.

## Task 6. Use BigQuery to analyze the audit logs

***Note:*** *Make sure you are on the* ***username 2:*** ***Google Cloud username 2*** *Google Cloud console.*

You've generated and exported logs to a BigQuery dataset. In this task, you'll analyze the logs using the Query editor.

***Note:** When you export logs to a BigQuery dataset, Cloud Logging creates dated tables to hold the exported log entries. Log entries are placed in tables whose names are based on the entries' log names.*

1. In the Google Cloud console, click the **Navigation menu** (☰).
2. Click **BigQuery**.

***Note:** The **Welcome to BigQuery in the Cloud Console** message box appears providing links to the quickstart guide and release notes for UI updates. Click **Done** to proceed.*

3. In the **Explorer** pane, click the expand arrow beside your project, Google Cloud project ID. The **auditlogs_dataset** dataset is displayed.

***Note:** If **auditlogs_dataset** is not displayed, reload your browser window.*

**Next**, verify that the BigQuery dataset has appropriate permissions to allow the export writer to store log entries.

4. Click the **auditlogs_dataset** dataset.

5. In the auditlogs_dataset toolbar, click the **Sharing** dropdown menu, and select **Permissions**.

6. On the **Share permission for "auditlogs_dataset"** page, expand the **BigQuery Data Editor** section.

7. Confirm that the service account used for log exports is a listed permission. The service account is similar to: service-xxxxxxxx@gcp-sa-logging.iam.gserviceaccount.com

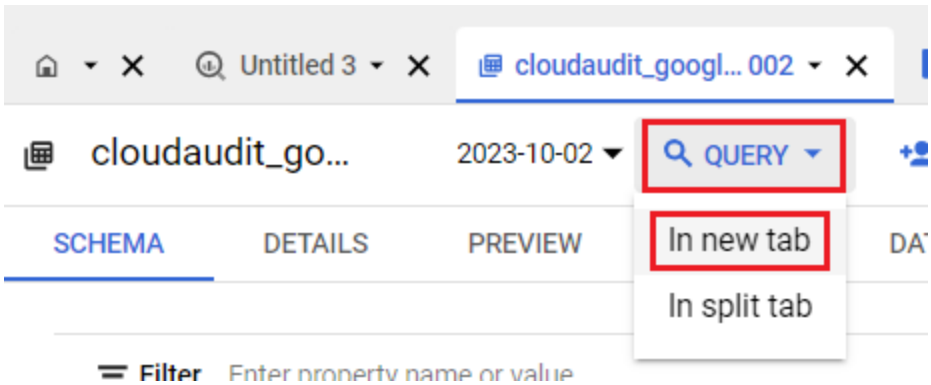
   This permission is assigned automatically when log exports are configured so this is a useful way to check that log exports have been configured.

8. Click **Close** to close the **Share Dataset** window.

9. In the **Explorer** pane, click the expander arrow next to the **auditlogs_dataset** dataset to view the **cloudaudit_googleapis_com_acitivty** table. This table contains your exported logs.

10. Select the **cloudaudit_googleapis_com_acitivty** table. The table schema displays. Take a moment to review the table schema and details.

11. Expand the **Query** drop-down menu and select **In new tab**.




12. In the Untitled tab of the query builder, delete any existing text and copy and paste the following command:

```
SELECT
  timestamp,
  resource.labels.instance_id,
  protopayload_auditlog.authenticationInfo.principalEmail,
  protopayload_auditlog.resourceName,
  protopayload_auditlog.methodName
FROM
`auditlogs_dataset.cloudaudit_googleapis_com_activity_*`
WHERE
  PARSE_DATE('%Y%m%d', _TABLE_SUFFIX) BETWEEN
  DATE_SUB(CURRENT_DATE(), INTERVAL 7 DAY) AND
  CURRENT_DATE()
  AND resource.type = "gce_instance"
  AND operation.first IS TRUE
  AND protopayload_auditlog.methodName = "v1.compute.instances.delete"
ORDER BY
  timestamp,
  resource.labels.instance_id
LIMIT
  1000;
```

Copied!

content_copy

This query returns the users that deleted virtual machines in the last 7 days.

13. Click **Run**.

After a couple of seconds, BigQuery will return each time a user deleted a Compute Engine virtual machine within the past 7 days. You should notice one entry, which is the activity you generated in the previous tasks as user 1. Remember, BigQuery shows only the activity that occurred after you created the export.

14. Replace the previous query in the **Untitled** tab with the following:

```
SELECT
  timestamp,
  resource.labels.bucket_name,
  protopayload_auditlog.authenticationInfo.principalEmail,
  protopayload_auditlog.resourceName,
  protopayload_auditlog.methodName
FROM
`auditlogs_dataset.cloudaudit_googleapis_com_activity_*`
WHERE
  PARSE_DATE('%Y%m%d', _TABLE_SUFFIX) BETWEEN
  DATE_SUB(CURRENT_DATE(), INTERVAL 7 DAY) AND
  CURRENT_DATE()
  AND resource.type = "gcs_bucket"
  AND protopayload_auditlog.methodName = "storage.buckets.delete"
ORDER BY
  timestamp,
  resource.labels.instance_id
LIMIT
  1000;
```

Copied!

content_copy

This query returns the users that deleted Cloud Storage buckets in the last 7 days. You should notice two entries, which is the activity you generated in the previous tasks as user 1.

15. Click **Run**.

The ability to analyze audit logs in BigQuery is very powerful. In this activity, you viewed just two examples of querying audit logs.

Click **Check my progress** to verify that you have completed this task correctly.

Use BigQuery to analyze the audit logs

Check my progress

**CONCLUSION**

Great work! You have successfully queried in Logs Explorer. You then exported logs and created a dataset that you analyzed in BigQuery.

You have shown how you can use audit logs and filter for types of malicious activity and then further analyze those logs in BigQuery as a way to analyze the threats.

| Exp. no: 11<br><br>Date: | Recover VMs with Google Backup and DR Service |
|---|---|

## How to start your lab and sign in to the Google Cloud console

1. Click the **Start Lab** button. On the left is the **Lab Details** panel with the following:

    - Time remaining
    - The **Open Google Cloud console** button
    - The temporary credentials that you must use for this lab
    - Other information, if needed, to step through this lab

    ***Note:*** *If you need to pay for the lab, a pop-up opens for you to select your payment method.*

2. Click **Open Google Cloud console** (or right-click and select **Open Link in Incognito Window**) if you are running the Chrome browser. The **Sign in** page opens in a new browser tab.

    ***Tip:*** You can arrange the tabs in separate, side-by-side windows to easily switch between them.

    ***Note:*** *If the* ***Choose an account*** *dialog displays, click* ***Use Another Account****.*

3. If necessary, copy the **Google Cloud username** below and paste it into the **Sign in** dialog. Click **Next**.

"Google Cloud username"
Copied!

content_copy

You can also find the **Google Cloud username** in the **Lab Details** panel.

4. Copy the **Google Cloud password** below and paste it into the **Welcome** dialog. Click **Next**.

"Google Cloud password"
Copied!

content_copy

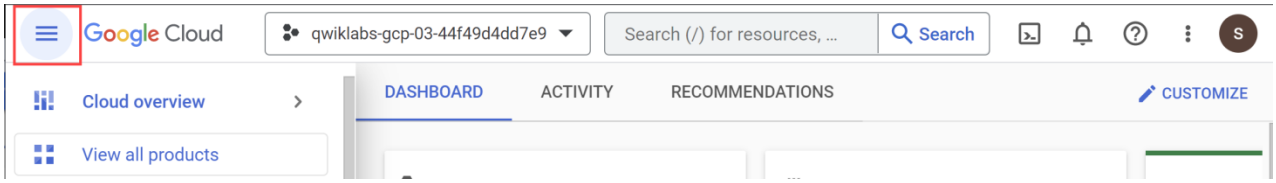
You can also find the **Google Cloud password** in the **Lab Details** panel.

***Important:*** *You must use the credentials the lab provides you. Do not use your Google Cloud account credentials.****Note:*** *Using your own Google Cloud account for this lab may incur extra charges.*

5. Click through the subsequent pages:
    - Accept the terms and conditions
    - Do not add recovery options or two-factor authentication (because this is a temporary account)
    - Do not sign up for free trials

After a few moments, the Console opens in this tab.

***Note:*** *You can view the menu with a list of Google Cloud Products and Services by clicking the* ***Navigation menu*** *at the top-left.*




## Task 1. Connect to the Backup and DR console

Before you can begin implementing recovery actions, you'll first need to connect to the Backup and DR console.

1. In the Google Cloud console, click the **Navigation menu** (☰) > **Backup and DR**. (You will have to click **More Products** and then scroll down to find **Backup and DR** in the **Operations** section).
2. From left nevigation pane, click **Management console**.
3. In the **Log in to the management console** section, click **Log in to the management console**.
4. If asked to **Choose an account**, click your **Google Cloud Username**: USERNAME.
5. **Skip** the **Welcome to Google Backup and DR!** tour. The Backup and DR management console opens.
6. In the Backup and DR management console titlebar, click **Manage** > **Appliances**.

If the management server and the Backup and Recovery server are successfully installed, the Connectivity status has a green check.

***Note:*** *If the* ***Update Status*** *is* ***Pending*** *(yellow exclamation point), an update is waiting for installation. You can ignore this and continue to your next task.*

**Task 2. Create a backup plan template**

In this task, you'll create a backup plan template.

Backup plan templates are composed of backup policies. In policies, you define when to run a backup, how frequently to run a backup, and how long to retain the backup image for — **Days**, **Weeks**, **Months**, or **Years**.

1. In the Backup and DR management console titlebar, click **Backup Plans** > **Templates**, and then click **+Create Template**.
2. In the **Template** field, set the template name to **vm-backup**.

***Note:*** *Template names are text strings. The only allowed special characters are spaces, underscores (_), and dashes (-).*

3. In the **Description** field, type **Virtual Machine Backups**.
4. In the **Policies** box, next to **Snapshot,** click **+ Add** to add a production to snapshot backup policy.

***Note:*** *If the **Policies** box is not displayed, scroll to the right or expand the browser window.*

The **Production to Snapshot** dialog opens.

5. In the **Create/Edit Policy** section, set the following fields and leave all other settings at their defaults:

| Field | Value |
|---|---|
| **Policy Name** | Daily VM snapshot |
| **Scheduling** | Continuous |
| **Every** | 2 Hour(s) |

***Note:*** *The **Scheduling** policy type can be either **Windowed** or **Continuous**. The default is **Windowed:***

• **Windowed** defines a discrete snapshot backup schedule adhering to a specific frequency and

time window.
• **Continuous** defines a continuous snapshot backup schedule

6. Click **Create Policy**.
7. Click **Save Template**.
8. Click **Okay** to acknowledge template creation **Success.**

Keep the Backup and DR management console open in a new tab for the entire lab.

Click **Check my progress** to verify that you have completed this task correctly.

Create a backup plan template

Check my progress

## Task 3. Validate the backup and recovery appliance service account permissions

In this task, you'll view the required IAM roles of the backup/recovery appliance to verify that it has the correct IAM roles.

An appliance is a hardware or software device that is designed to perform a specific task. Security appliances are used to protect networks from unauthorized access, attacks, and data breaches.

Every appliance has a dedicated service account attached to it—that was created during appliance deployment in the project where the appliance was deployed. For appliances installed on version 11.0.2 and higher, a corresponding cloud credential for this service account is automatically created at the time of an appliance deployment.

The name of the cloud credential is based on the appliance name followed by the suffix -sa. For example, if the name of the backup/recovery appliance is bur-appliance-us-east1, then the name appliances corresponding cloud credential is bur-appliance-us-east1-sa.

To view and verify the required IAM role:

1. Return to the Google Cloud console, in the **Navigation menu** (☰), click **IAM & Admin** > **IAM**.
2. In the **Name** column, find the service account attached to your backup appliance, the service account's name should be **Service account for backup and recovery appliance**.
3. In the **Role** column, notice that the **Backup and DR Cloud Storage Operator role** is already assigned.

**Task 4. Discover and add Compute Engine instances to the management console**

In this task, you'll use the **onboarding wizard** to onboard your Compute Engine instances. Onboarding an instance means you attach the template to the instance.

1. Return to the Backup and DR management console.
2. From the titlebar, click **Backup and Recover** > **Back Up**.
3. In the **Google Cloud** section, click **Compute Engine**.
4. Under **Credential**, select **backup**, and click **Next**.

The **Project ID** and **Zone** drop-down options are populated with details from the appliance that maps to the **workflow** credential.

5. Click **Search**.

The results are listed in the search results. You may have to scroll down to view them:

- **lab-vm**
- **qwiklabs-appliance**

6. Select the **lab-vm** Compute Engine instance for backup, and then click **Next**.

***Note:*** *If no instances or only one instance appear, ensure that the zone selected matches the zone where your Compute Engine instance(lab-vm) is located or running.*

7. In the **Enable backups for Compute Engine VM instances?** page, select the **lab-vm** and then set the following:
   - **Action:** From the drop-down menu, select **Apply a backup template**.
   - **Backup template:** From the drop-down menu, select **vm-backup**.
8. Click **OK**.
9. Click **Next**.

A **Summary of changes** screen appears and provides the following information:

- Instance Name: lab-vm
- Appliance: qwiklabs-appliance
- Action: Apply a backup template

11. Click **Finish** to complete the onboarding process. This triggers the back up of the selected Compute Engine instances based on the **Policy Template** you attached.
12. Click **Finish** to confirm your intent to finish.

After onboarding is complete the **Status** is a green check. This means the policy template is attached to the selected VM.

***Note:*** *Backup and DR ensures that the chosen Compute Engine instances get backed up at the frequency you set in the backup policy.*

13. In the Backup and DR management console titlebar, click **Monitor** > **Jobs**.

You can monitor the progress of the backup job. When the job is finished, you have an image that you can restore if needed.

If the jobs list is empty, the backup job has either not started or is already completed. Use different filter options to populate the jobs list, for example **Succeeded** or **All** filter options. Filter results are listed in the **Jobs** list.

***Note:*** *The job may take five minutes or longer to finish.*

Click **Check my progress** to verify that you have completed this task correctly.

Discover and add Compute Engine instances to the management console

Check my progress

**Task 5. Restore a Compute Engine instance**

Now that you have an image of your Compute Engine instance, in this task, you'll create a brand new Compute Engine instance using the backup image that you created in the previous task.

1. From the Backup and DR management console titlebar, click **Backup & Recover** > **Recover**.
2. Click the name of the Compute Engine instance you want to recover (**lab-vm**) to select it. Click **Next**.
3. In the action bar, click **Table**. In the **Images** list, one image is displayed because there has only been one backup image created.
4. Select the image and click **Mount**.

***Note:*** *Typically, the **Mount** panel has many selection choices that allow you to choose where and how to restore an image. In this lab, you may have only one timeline option as you just created the first backup.*

5. Under **Mount**, select **Mount as new GCE instance**.
6. Review the configuration options and then update the following:
    - **Region**: Change this to REGION
    - **Zone**: Change this to ZONE
    - **Instance name**: lab-vm-recovered
7. Scroll to the bottom of the page and click **Mount**.
8. On the **Success** dialog, click **Go to Job Monitor**.
9. In the filter pane, in the **Status** section, uncheck **Running**. Two jobs are displayed, an earlier with a **Succeeded** status, and the one you just started with a **Running** status.

When both jobs have a **Succeeded** status, you have the Computer Engine instance.

***Note:*** *The job may take five minutes or longer depending on the region you selected.*

To view the recovered VM, go to the Google Cloud console, in **Navigation menu** (☰), click **Compute Engine** > **VM instances** to view three VM instances:

- **lab-vm**
- **lab-vm-recovered**
- **qwiklabs-appliance**

Click **Check my progress** to verify that you have completed this task correctly.

Restore a Compute Engine instance

Check my progress

## Task 6. Restore a Compute Engine instance to an alternate project

In this task, you'll restore a Computer Engine instance using the back up template you created, but this time to a different project.

You can also create a brand new Compute Engine instance in a different project from backup images.

***Note:** Before you set the default service account as a **Principal** in a different project, you must add the default service account as a **Principal** in the target project.*

To restore a Compute Engine instance to an alternate project, you first add the service account of project 1 as a principle to Google Cloud project 2 and then recover the instance on Google Cloud project 2:

1. In the Google Cloud console, in the Navigation menu (☰), click **IAM & Admin** > **IAM**.
2. In the list of principals, find and copy the email of the **Service account for backup and recovery appliance** to use in Step 6. The email is similar to the following: *qwiklabs-appliances@qwiklabs-gcp-xx-xxxxxxxxx.iam.gserviceaccount.com*.
3. In the Google Cloud console, click the **Project** selection drop-down. If the project lists only one project, click **All** to open the **All** tab.
4. Search for Google Cloud project ID 2:PROJECT_ID_2 and then click to select that project ID. You are now in the **Permissions** page for Google Cloud project ID 2:PROJECT_ID_2.
5. Click **Grant access**.
6. In the **Add Principles** section, in the **New principals** field, paste the email address of the service account of Google Cloud project 1, named **Service account for backup and recovery appliance**. It should be still in your clipboard.
7. In the **Assign roles** section:
    - Click **Select a role** and assign the **Backup and DR** > **Backup and DR Compute Engine Operator** role.
    - Click **+Add Another Role**.
    - Click **Select a role** and assign the **Backup and DR** > **Backup and DR Cloud Storage Operator** role.
8. Click **Save**.

You've added the service account of Google Cloud project 1 as a principal to Google Cloud project 2. You can now recover the instance on google Cloud project 2.

9. From the Backup and DR management console, navigate to **Backup & Recover** > **Recover**.
10. Select **lab-vm** (the Compute Engine instance you want to recover), and click **Next**.

***Note:*** *The green check next to the instance means that instance has been backed up, the red X means it hasn't.*

11. In the action bar, click **Table.**
12. In the **Images** list, select the top image and then click **Mount**.
13. Under **Mount**, select **Mount as new GCE instance**.
14. Review the configuration and update the following options:
    - **Project**: Change this to PROJECT_ID_2 to simulate recovering to a different project in the Google Cloud.
    - **Instance name**: Notice you can use the same instance name because you are in a different Google Cloud project. Update the instance name to **lab-vm-project2**.
    - **Region**: Change this to REGION.
    - **Zone**: Change this to ZONE.
15. Select **Mount** at the bottom of the panel. A **Mount** job starts. On the **Success** dialog, click **Go to Job Monitor** to monitor the status of this current job. The job may take five minutes or longer depending on what region you selected.
16. To view the recovered Compute Engine instance in the Google Cloud console of Google Cloud project 2, in the **Navigation menu**, click **Compute Engine > VM instances**.

***Note:*** *The job may take five minutes or longer to finish.****Note:*** *Before you set the default service account as a* ***Principal*** *in a different project, you must add the default service account as a* ***Principal*** *in the target project.*

Click **Check my progress** to verify that you have completed this task correctly.

Restore a Compute Engine instance to an alternate project

Check my progress

## Conclusion

Great work! You successfully used Google Backup and DR Service to create a backup template and then applied it to two Compute Engine instances.

You have shown how to prepare for issues with VMs and the service. When a device malfunctions, you can use Backup and DR Service to restore mal-functioning devices across multiple Google Cloud projects.

| Exp. no: 12<br><br>Date: | Respond and recover from a data breach |
|---|---|

## How to start your lab and sign in to the Google Cloud console

1. Click the **Start Lab** button. On the left is the **Lab Details** panel with the following:

    - Time remaining
    - The **Open Google Cloud console** button
    - The temporary credentials that you must use for this lab
    - Other information, if needed, to step through this lab

    ***Note:*** *If you need to pay for the lab, a pop-up opens for you to select your payment method.*

2. Click **Open Google Cloud console** (or right-click and select **Open Link in Incognito Window**) if you are running the Chrome browser. The **Sign in** page opens in a new browser tab.

    ***Tip:*** You can arrange the tabs in separate, side-by-side windows to easily switch between them.

    ***Note:*** *If the* ***Choose an account*** *dialog displays, click* ***Use Another Account****.*

3. If necessary, copy the **Google Cloud username** below and paste it into the **Sign in** dialog. Click **Next**.

"Google Cloud username"
Copied!

content_copy

You can also find the **Google Cloud username** in the **Lab Details** panel.

4. Copy the **Google Cloud password** below and paste it into the **Welcome** dialog. Click **Next**.

"Google Cloud password"
Copied!

content_copy

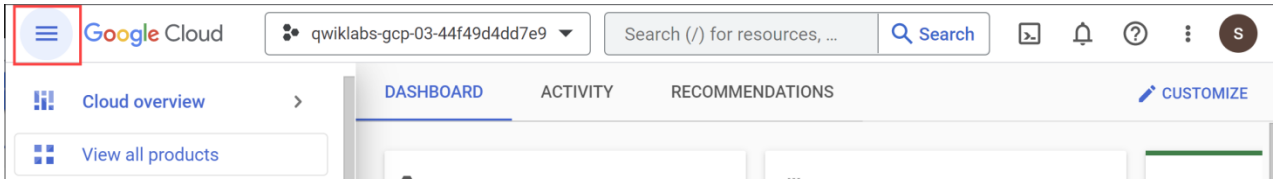
You can also find the **Google Cloud password** in the **Lab Details** panel.

***Important:*** *You must use the credentials the lab provides you. Do not use your Google Cloud account credentials.****Note:*** *Using your own Google Cloud account for this lab may incur extra charges.*

5. Click through the subsequent pages:
    - Accept the terms and conditions
    - Do not add recovery options or two-factor authentication (because this is a temporary account)
    - Do not sign up for free trials

After a few moments, the Console opens in this tab.

***Note:*** *You can view the menu with a list of Google Cloud Products and Services by clicking the* ***Navigation menu*** *at the top-left.*




## Task 1. Analyze the data breach and gather information

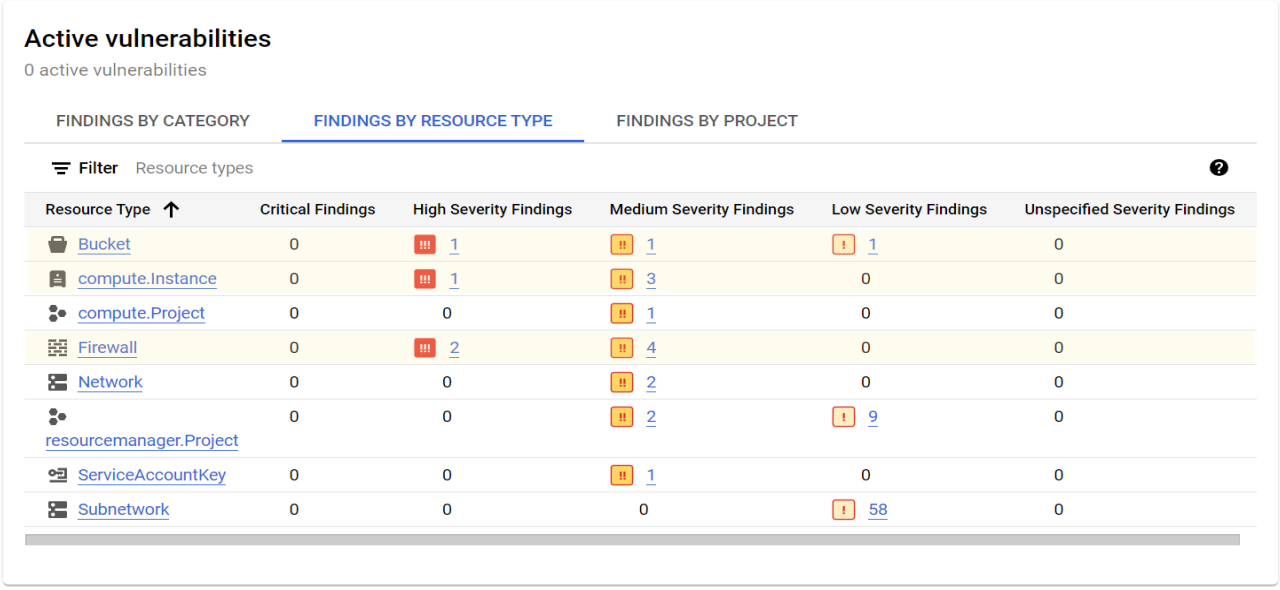
One morning, the security team detects unusual activity within their systems. Further investigation into this activity quickly reveals that the company has suffered a massive security breach across its applications, networks, systems, and data repositories. Attackers gained unauthorized access to sensitive customer information, including credit card data, and personal details. This incident requires immediate attention and thorough investigation. The first step towards understanding the scope and impact of this breach is to gather information and analyze the available data.

In this task, you'll examine the vulnerabilities and findings in Google Cloud Security Command Center to determine how the attackers gained access to the data, and which remediation steps to take.

***Important:*** *The vulnerabilities listed in this section rely on specific security checks being run beforehand. If some checks haven't run yet, the related vulnerabilities might not appear in the Security Command Center when you complete the steps in this section. Don't worry though! You can still use the information provided in this task to analyze the available findings and proceed with the remediation steps in the tasks that follow.*

**First**, navigate to the Security Command Center to view an overview of the active vulnerabilities.

1. In the Google Cloud console, in the **Navigation menu** (☰), click **Security** > **Risk Overview**. The Security Command Center Overview page opens.
2. Scroll down to **Active vulnerabilities**. This provides an overview of current security vulnerabilities or issues that need attention within the Google Cloud environment.
3. Select the **Findings By Resource Type** tab. The security findings or vulnerabilities based on the type of cloud resource affected (e.g., instances, buckets, databases) are organized. By reviewing active vulnerabilities and findings by resource type, you can prioritize and address security issues effectively.




| Resource Type | Critical Findings | High Severity Findings | Medium Severity Findings | Low Severity Findings | Unspecified Severity Findings |
|---|---|---|---|---|---|
| Bucket | 0 | 1 | 1 | 1 | 0 |
| compute.Instance | 0 | 1 | 3 | 0 | 0 |
| compute.Project | 0 | 0 | 1 | 0 | 0 |
| Firewall | 0 | 2 | 4 | 0 | 0 |
| Network | 0 | 0 | 2 | 0 | 0 |
| resourcemanager.Project | 0 | 0 | 2 | 9 | 0 |
| ServiceAccountKey | 0 | 0 | 1 | 0 | 0 |
| Subnetwork | 0 | 0 | 0 | 58 | 0 |

You'll note that there are both high and medium severity findings relating to the **Cloud Storage bucket**, the **Compute Instance virtual machine**, and the **firewall**.

Which three resource types are listed with high severity findings?

Bucket, Subnetwork, and ServiceAccountKey

Bucket, compute.Instance, and Firewall

Network, Subnetwork, and compute.Instance

Network, Firewall, and Bucket

Submit

**Next**, navigate to the PCI DSS report.

4. In the **Security Command Center** menu, click **Compliance**. The Compliance page opens.
5. In the **Google Cloud compliance standards** section, click **View details** in the **PCI DSS 3.2.1** tile. The PCI DSS 3.2.1 report opens.

6. Click on the **Findings** column to sort the findings and display the active findings at the top of the list.

***Note:*** *Make sure to follow these steps to assess the PCI report, and do not refresh the page, as the required filters will be removed, and the correct information won't be displayed.*

The Payment Card Industry Data Security Standard (PCI DSS) is a set of security requirements that organizations must follow to protect sensitive cardholder data. As a retail company that accepts and processes credit card payments, Cymbal Retail must also ensure compliance with the PCI DSS requirements, to protect cardholder data.

As you examine the PCI DSS 3.2.1 report, notice that it lists the rules that are non-compliant, which relate to the data breach:

- **Firewall rule logging should be enabled so you can audit network access**: This medium severity finding indicates that firewall rule logging is disabled, meaning that there is no record of which firewall rules are being applied and what traffic is being allowed or denied. This is a security risk as it makes it difficult to track and investigate suspicious activity.
- **Firewall rules should not allow connections from all IP addresses on TCP or UDP port 3389**: This high severity finding indicates that the firewall is configured to allow Remote Desktop Protocol (RDP) traffic for all instances in the network from the whole internet. This is a security risk as it allows anyone on the internet to connect to the RDP port on any instance in the network.
- **Firewall rules should not allow connections from all IP addresses on TCP or SCTP port 22**: This high severity finding indicates that the firewall is configured to allow Secure Shell (SSH) traffic to all instances in the network from the whole internet. SSH is a protocol that allows secure remote access to a computer. If an attacker can gain access to a machine through SSH, they could potentially steal data, install malware, or disrupt operations.
- **VMs should not be assigned public IP addresses**: This high severity finding indicates that a particular IP address is actively exposed to the public internet and is potentially accessible to unauthorized individuals. This finding is considered a potential security risk because it could allow attackers to scan for vulnerabilities or launch attacks on the associated resource.
- **Cloud Storage buckets should not be anonymously or publicly accessible**: This high severity finding indicates that there is an Access Control List (ACL) entry for the storage bucket that is publicly accessible which means that anyone on the internet can read files stored in the bucket. This is a high-risk security vulnerability that needs to be prioritized for remediation.
- **Instances should not be configured to use the default service account with full access to all Cloud APIs**: This medium severity finding indicates that a particular identity or service account has been granted full access to all Google Cloud APIs. This finding is considered a significant security risk because it grants the identity or service account the ability to perform any action within the Google Cloud environment, including accessing sensitive data, modifying configurations, and deleting resources.

  Since you're focusing on identifying and remediating the issues related to the security incident, please disregard the following findings as they do not relate to the remediation tasks you're completing:

- **VPC Flow logs should be Enabled for every subnet VPC Network**: There are a number of low severity findings for Flow Logs disabled. This indicates that Flow Logs are not enabled for a number of subnetworks in the Google Cloud project used for this lab. This is a potential security risk because Flow Logs provide valuable insights into network traffic patterns, which can help identify suspicious activity and investigate security incidents.
  ***Note:*** *Enabling logging for cloud resources is important in maintaining observability. However, you will not remediate this finding in this lab activity as the subnetworks are part of this lab environment. As a result, this finding will still be visible on the report after you have completed the remediation tasks.*

- **Basic roles (Owner, Writer, Reader) are too permissive and should not be used**: This medium severity finding indicates that primitive roles are being used within the Google Cloud environment. This is a potential security risk because primitive roles grant broad access to a wide range of resources.
- **An egress deny rule should be set**: This low severity finding indicates that no egress deny rule is defined for the monitored firewall. This finding raises potential security concerns because it suggests that outbound traffic is not restricted, potentially exposing sensitive data or allowing unauthorized communication.
  The following table pairs the rules listed in the report with their corresponding findings category. This will assist you when examining the findings according to resource type later:

| Findings category | Rule |
|---|---|
| Firewall rule logging disabled | Firewall rule logging should be enabled so you can audit network access |
| Open RDP port | Firewall rules should not allow connections from all IP addresses on TCP or UDP port 3389 |
| Open SSH port | Firewall rules should not allow connections from all IP addresses on TCP or SCTP port 22 |
| Public IP address | VMs should not be assigned public IP addresses |

| | |
|---|---|
| Public bucket ACL | Cloud Storage buckets should not be anonymously or publicly accessible |
| Full API access | Instances should not be configured to use the default service account with full access to all Cloud APIs |
| Flow logs disabled | VPC Flow logs should be Enabled for every subnet VPC Network |
| Primitive roles used | Basic roles (Owner, Writer, Reader) are too permissive and should not be used |
| Egress deny rule not set | An egress deny rule should be set |

Overall, these findings indicate a critical lack of security controls and non-compliance with essential PCI DSS requirements; they also point to the vulnerabilities associated with the data breach.

**Next**, navigate to the Security Command Center, and filter the findings for further examination and analysis of the vulnerabilities in the Google Cloud environment.

7. In the Google Cloud console, in the **Navigation menu** (☰), click **Security** > **Findings**. The **Findings** page opens.
8. In the **Quick filters** panel, in the **Resource Type** section, select the checkbox for the **Google Cloud storage bucket** resource type.

The following active findings pertaining to the storage bucket should be listed:

- **Public bucket ACL**: This finding is listed in the PCI DSS report, and indicates that anyone with access to the internet can read the data stored in the bucket.
- **Bucket policy only disabled**: This indicates that there is no explicit bucket policy in place to control who can access the data in the bucket.
- **Bucket logging disabled**: This indicates that there is no logging enabled for the bucket, so it will be difficult to track who is accessing the data.

These findings indicate that the bucket is configured with a combination of security settings that could expose the data to unauthorized access. You'll need to remediate these findings by

removing the public access control list, disabling public bucket access, and enabling the uniform bucket level access policy.

***Note:** Enabling logging for cloud resources is important in maintaining observability. However, you will not remediate the Bucket logging disabled finding in this lab activity as this would require working with multiple projects. As a result, this finding will still be visible after you have completed the remediation tasks.*

9. In the **Quick filters** panel, in the **Resource Type** section, uncheck **Google Cloud storage bucket,** andselect the checkbox for the **Google compute instance** resource type.

The following active findings that pertain to the virtual machine named **cc-app-01** should be listed:

- **Malware bad domain**: This finding indicates that a domain known to be associated with malware was accessed from the google.compute.instance named cc-app-01. Although this finding is considered to be of low severity, it indicates that malicious activity has occurred on the virtual machine instance and that it has been compromised.
- **Compute secure boot disabled**: This medium severity finding indicates that secure boot is disabled for the virtual machine. This is a security risk as it allows the virtual machine to boot with unauthorized code, which could be used to compromise the system.
- **Default service account used**: This medium severity finding indicates that the virtual machine is using the default service account. This is a security risk as the default service account has a high level of access and could be compromised if an attacker gains access to the project.
- **Public IP address**: This high severity finding is listed in the PCI DSS report and indicates that the virtual machine has a public IP address. This is a security risk as it allows anyone on the internet to connect to the virtual machine directly.
- **Full API access**: This medium severity finding is listed in the PCI DSS report, and indicates that the virtual machine has been granted full access to all Google Cloud APIs.

These findings indicate the virtual machine was configured in a way that left it very vulnerable to the attack. To remediate these findings you'll shut the original VM (cc-app-01) down, and create a VM (cc-app-02) using a clean snapshot of the disk. The new VM will have the following settings in place:

- No compute service account
- Firewall rule tag for a new rule for controlled SSH access
- Secure boot enabled
- Public IP address set to None

10. In the **Time range** field, expand the drop-down, and select **Last 30 days**. This will ensure the list includes findings for the last 30 days.
11. In the **Quick filters** panel, in the **Resource Type** section, uncheck **Google compute instance**, and select the checkbox for the **Google compute firewall** resource type.

The following active findings should be listed that pertain to the firewall:

- **Open SSH port**: This high severity finding indicates that the firewall is configured to allow Secure Shell (SSH) traffic to all instances in the network from the whole internet.
- **Open RDP port**: This high severity finding indicates that the firewall is configured to allow Remote Desktop Protocol (RDP) traffic to all instances in the network from the whole internet.
- **Firewall rule logging disabled**: This medium severity finding indicates that firewall rule logging is disabled. This means that there is no record of which firewall rules are being applied and what traffic is being allowed or denied.

These findings are all listed in the PCI DSS report and highlight a significant security gap in the network's configuration. The lack of restricted access to RDP and SSH ports, coupled with disabled firewall rule logging, makes the network highly vulnerable to unauthorized access attempts and potential data breaches. You'll need to remediate these by removing the existing firewall overly broad rules, and replacing them with a firewall rule that allows SSH access only from the addresses that are used by Google Cloud's IAP SSH service.

Now that you have analyzed the security vulnerabilities, it’s time to work on remediating the report findings.

Which of the following findings are listed as high severity findings?

Public bucket ACL, Public IP address, Open SSH port, and Open RDP port

Public IP address, Default service account used, Full API access, and Firewall rule logging disabled

Bucket policy only disabled, Bucket logging disabled, Malware bad domain, and Compute secure boot disabled

Firewall rule logging disabled, Compute secure boot disabled, Public IP address, and Bucket logging disabled

Submit

**Task 2. Fix the Compute Engine vulnerabilities**

In this task, you'll shut down the vulnerable VM **cc-app-01**, and create a new VM from a snapshot taken before the malware infection. VM snapshots are effective in restoring the system to a clean state, and ensures that the new VM will not be infected with the same malware that compromised the original VM.

1. In the Google Cloud console, click the **Navigation menu** (☰).
2. Select **Compute Engine > VM instances**. The VM instances page opens.

The current VM **cc-app-01** should be listed under VM instances. This is the vulnerable VM that has been compromised and must be shut down.

3. Select the checkbox for the **cc-app-01** VM.
4. Click **Stop**.

5. A pop-up will appear asking you to confirm that the VM should be stopped, click **Stop**.

Click **Check my progress** to verify that you have completed this task correctly.

Shut down the vulnerable VM

Check my progress

**Next**, create a new VM from a snapshot. This snapshot has already been created as part of Cymbal Retail's long term data backup plan.

6. In the action bar, click **+ Create instance**.
7. In the **Name** field, type **cc-app-02**.
8. In the **Machine type** section, expand the drop-down, select **Shared-core**, and then select **e2-medium**.
9. Click **OS and storage**.
10. In the **Operating system and storage** section, click **Change**. The Boot disk dialog opens.
11. Select the **Snapshots** tab.
12. Expand the **Snapshot** drop-down menu, and select **cc-app01-snapshot**.
13. Click **Select**.
14. Click **Security**.
15. In the **Identity and API access** section, expand the **Service accounts** drop-down menu, and select **Qwiklabs User Service Account**.
16. Click **Networking**.
17. In the **Network tags** field, type **cc**. You'll use this tag to apply firewall rules to this specific VM.
18. In the **Network interfaces** section, expand the **default** network.
19. Expand the **External IPv4 address** drop-down menu, and select **None**.
20. Click **Create**.

The new VM **cc-app-02** should now be created from the **cc-app01-snapshot.** (It may take a few minutes for the new VM to be created.)

**Now**, turn Secure Boot on for the new VM **cc-app-02** to address the **Secure Boot disabled** finding.

20. Select the checkbox for the **cc-app-02** VM.
21. Click **Stop**.
22. A pop-up will appear asking you to confirm that the VM should be stopped, click **Stop**.

Wait for the **cc-app-02** VM to be stopped before you continue.

23. In the **VM instances** section, click the **cc-app-02** link. The cc-app-02 page opens.
24. In the **cc-app-02** toolbar, click **Edit.** The Edit cc-app-02 instance page opens.
25. Scroll down to the **Security and access** section, and under **Shielded VM**, select the checkbox for the **Turn on Secure Boot** option. This will address the **Compute secure boot disabled** finding.
26. Click **Save**.

27. In the **Compute Engine** menu, select **VM instances**.
28. Select the checkbox for the **cc-app-02** VM.
29. Click **Start/Resume**.
30. A pop-up will appear asking you to confirm that the VM should be started, click **Start**.

The **cc-app-02** VM instance will restart and the **Secure Boot disabled** finding will be remediated.

Click **Check my progress** to verify that you have completed this task correctly.

Create a new VM from existing snapshot

Check my progress

Challenge: Delete the compromised VM

Delete the compromised VM **cc-app-01**.

Click **Check my progress** to verify that you have completed this task correctly.

Delete the compromised VM

Check my progress

By following these steps, you have effectively created a new VM from the snapshot, ensuring it is free from malware and misconfigurations. You also deleted the compromised VM, eliminating the source of the security breach.

**Task 3. Fix Cloud Storage bucket permissions**

In this task, you'll revoke public access to the storage bucket and switch to uniform bucket-level access control, significantly reducing the risk of data breaches. By removing all user permissions from the storage bucket, you can prevent unauthorized access to the data stored within.

1. In the **Navigation menu** (☰), select **Cloud Storage > Buckets**. The Buckets page opens.
2. Click the **project_id_bucket** storage bucket link. The Bucket details page opens.

You'll note there is a **myfile.csv** file in the publicly accessible bucket. This is the file that contains the sensitive information that was dumped by the malicious actor. Perform the following steps to address the **Public bucket ACL** finding.

3. Click the **Permissions** tab.

4. In the **Public access** tile, click **Prevent public access.**

5. Click **Confirm**.

Challenge: Modify storage bucket access

Switch the access control to uniform and remove permissions for the **allUsers** principals from the storage bucket to enforce a single set of permissions for the bucket and its objects. You'll also need to ensure that users who rely on basic project roles to access the bucket won't lose their access.

Click **Check my progress** to verify that you have completed this task correctly.

Modify storage bucket access.

Check my progress

By following these steps, you have effectively prevented public access to the bucket, switched to uniform bucket-level access control, and removed all user permissions, addressing the **Public bucket ACL**, **Bucket policy only disabled**, and **Bucket logging disabled** findings.

**Task 4. Limit firewall ports access**

In this task, you'll restrict access to RDP and SSH ports to only authorized source networks to minimize the attack surface and reduce the risk of unauthorized remote access.

Exercise extreme caution before modifying overly permissive firewall rules. The rules may be allowing legitimate traffic, and improperly restricting it could disrupt critical operations. In this lab, ensure the Compute Engine virtual machine instances tagged with target tag "cc" remain accessible via SSH connections from the Google Cloud Identity-Aware Proxy address range (35.235.240.0/20). To maintain uninterrupted management access, create a new, limited-access firewall rule for SSH traffic before removing the existing rule allowing SSH connections from any address.

**Challenge: Restrict SSH access**

Create a new firewall rule. This rule must restrict SSH access to only authorized IP addresses from the source network **35.235.240.0/20** to compute instances with the target tag **cc**.

Click **Check my progress** to verify that you have completed this task correctly.

Restrict SSH access

Check my progress

**Task 5. Fix the firewall configuration**

In this task, you'll delete three specific VPC firewall rules that are responsible for allowing unrestricted access to certain network protocols, namely ICMP, RDP, and SSH, from any source within the VPC network. Then, you'll enable logging on the remaining firewall rules.

**Challenge: Customize firewall rules**

Delete the **default-allow-icmp**, **default-allow-rdp**, and **default-allow-ssh** firewall rules. These rules are overly broad and by deleting them, you'll allow for a more secure and controlled network environment.

By deleting these rules, you have restricted access to these protocols, limiting the potential for unauthorized access attempts and reducing the attack surface of your network.

Customize firewall rules

Check my progress

**Challenge: Enable logging**

Enable logging for the remaining firewall rules **limit-ports** (the rule you created in a previous task) and **default-allow-internal**.

Enabling logging allows you to track and analyze the traffic that is allowed by this rule, which is likely to be internal traffic between instances within your VPC.

Click **Check my progress** to verify that you have completed this task correctly.

Enable logging

Check my progress

By customizing firewall rules and enabling logging, you've addressed the **Open SSH port**, **Open RDP port**, and **Firewall rule logging disabled** findings. The new firewall rule better protects the network and improves network visibility.

**Task 6. Verify compliance**

After diligently addressing the vulnerabilities identified in the PCI DSS 3.2.1 report, it's crucial to verify the effectiveness of your remediation efforts. In this task, you'll run the report again to

ensure that the previously identified vulnerabilities have been successfully mitigated and no longer pose a security risk to the environment.

1. In the **Security Command Center** menu, click **Compliance**. The Compliance page opens.
2. In the **Google Cloud compliance standards** section, click **View details** in the **PCI DSS 3.2.1** tile. The PCI DSS 3.2.1 report opens.
3. Click on the **Findings** column to sort the findings and display the active findings at the top of the list.

All major vulnerabilities are now resolved.

***Note:*** *While you addressed the high and medium severity vulnerabilities, the flow logs remain disabled for a number of subnetworks. This finding will still be visible on the report after you have completed the remediation tasks, as this relates to this lab environment.*

**CONCLUSION**

You have helped the security team at Cymbal Bank to mitigate the impact of the data breach, address the identified vulnerabilities, and significantly enhanced the security posture of Cymbal Bank's Google Cloud environment.

First, you examined and analyzed the vulnerabilities and findings in Google Cloud Security Command Centre.

Next, you shut the old VM down and created a new VM from a snapshot taken before the malware infection.

Then, you fixed the cloud storage permissions by revoking public access to the storage bucket and switching to uniform bucket-level access control. You also removed all user permissions from the storage bucket.

Next, you fixed the firewall rules by deleting the default-allow-icmp, default-allow-rdp, and default-allow-ssh firewall rules, and enabling logging for the remaining firewall rules.

Finally, you run a compliance report to confirm that the vulnerability issues have been remediated.

Remember, as a security analyst it is crucial to maintain regular security audits and implement ongoing monitoring practices for continued protection against evolving threats and vulnerabilities.